A CPS-free proof of $BB(2) = 6$ in $I\Delta_0$

Loop certificates, one bounded induction, and finite coverage

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Abstract

A two-state machine either halts within six steps, or its proposed halting histories are excluded by a finite tape-pattern invariant. We give the common arithmetic proofs as exact natural-deduction tables, including the single bounded-induction instance used for history propagation. The invariants are certificates for ordinary cyclers and translated cyclers; no n-gram or closed-position-set (CPS) argument is used. A companion Metamath database supplies the literal finite instantiations and a six-step witness. The distinction between the common soundness theorem, the finite certificate checks, and the final numerical value is maintained throughout.

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1 What has to be proved, and what "8" means

Here $BB(2)$ is the largest number of executed instructions before halting, among deterministic machines with two working states, two tape symbols, a two-way infinite tape initially blank, and a separate halting state. The target is $BB(2) = 6$, not the maximum number of printed ones.

A *cycler* returns to a previous complete configuration. A *translated cycler* repeats its local behavior while moving through the blank part of the tape, leaving behind information that will never again be read. Ordinary simulation can find such a repetition; a proof must explain why it continues to prevent halting at every later time.

The previously discussed parameter 8 is a loop-search budget. It is not the number of steps before which every machine must halt. A finite reproduction of the Rocq `Loops` test accepts every nonhalting entry of its published 61-entry BB2 list with budget 8; budget 7 misses entries. We do not identify that reproduction with a formal proof of the entire Rocq program. Instead we use a simpler proof-producing certificate checker. Its certificates are found by time 5 for those 42 nonhalting entries, and contain at most five distinct rows. We pad to eight rows by repeating a row, which changes no represented set.

The formal soundness theorem below applies to *any* eight-row chart whose initial-membership, one-step-closure and final-state conditions have been proved. The three pattern lemmas explain how the checker proves those conditions by finite arithmetic. Thus a search is never trusted merely because it reports "loop found." It must supply a certificate and an actual proof of closure. This is a new certificate-based formulation of the loop tests, not a verification of the literal Rocq implementation.

Reading guide and proof boundary. Sections 2–5 give the definitions and explain the construction. Section 8 prints every natural-deduction line of the 30 shared proofs (958 lines). The finite, machine-specific instantiations are supplied as full natural-deduction traces and Metamath proofs in the source archive, rather than repeatedly printing the same proof pattern. The numerical lower-bound witness is also supplied there. The PDF therefore prints the complete *general soundness core*; the companion files are part of the complete formal BB(2) proof, and are needed to inspect its expanded finite instantiations. No unproved arithmetic lemma is added as an axiom.

2 The arithmetic and its proof rules

The language has $0, 1, +, \cdot, <, =$ and the usual first-order logical symbols. Numerals such as 2 abbreviate fixed terms: $2 = (1 + 1)$ and $3 = ((1 + 1) + 1)$. Products and sums keep their displayed parentheses; reassociating them is an arithmetic proof step.

All free variables in the following axioms may be replaced by terms, and each instance may be used under any open assumptions. The axioms are the ordered-semiring presentation used by the existing machine-history formalization.

A1	$((t + u) + v) = t + (u + v)$
A2	$t + u = u + t$
A3	$t + 0 = t$
A4	$((t \cdot u) \cdot v) = t \cdot (u \cdot v)$

A5	$t \cdot u = u \cdot t$
A6	$t \cdot 1 = t$
A7	$t \cdot 0 = 0$
A8	$t \cdot (u + v) = (t \cdot u) + (t \cdot v)$
O1	$\neg(t < t)$
O2	$((t < u) \wedge (u < v)) \to (t < v)$
O3	$(t < u) \vee ((t = u) \vee (u < t))$
O4	$\neg(t < 0)$
O5	$(t < u) \to (t + v < u + v)$
O6	$((t < u) \wedge (0 < v)) \to (t \cdot v < u \cdot v)$
O7	$(t < u) \leftrightarrow \exists x\,[u = ((t + x) + 1)]$, with x absent from t, u.

A bounded quantifier is a textual abbreviation:

$$\forall x < t\, P := \forall x(x < t \to P), \qquad \exists x < t\, P := \exists x(x < t \wedge P).$$

A formula is Δ_0 if it is built from atomic equalities and inequalities using logical connectives and bounded quantifiers; the bound on a quantified variable must not contain that variable. Parameters are allowed. The induction schema is

$$\big(\varphi(0) \wedge \forall i(\varphi(i) \to \varphi(i+1))\big) \to \forall i \varphi(i), \quad \varphi \in \Delta_0.$$

The quantifier $\forall i$ in the induction *axiom* is unbounded. The restriction concerns the formula φ to which induction is applied, not the quantifiers used in ordinary reasoning.

A table line has a list of undischarged assumptions, a formula, one rule, and references to its premises. Assume introduces the last assumption; Weak keeps a result after adding an assumption; Closed-Weak brings an already closed result into a context. The connective rules are the usual natural-deduction introductions and eliminations. In particular $\vee E$ cites a disjunction and both subderivations, and $\to I$ discharges the last assumption. $\neg I$ derives $\neg P$ from derivations of Q and $\neg Q$ under an additional assumption P; $\bot E$ derives any formula from such a contradictory pair. Classical excluded middle and double-negation elimination are available in the kernel; the common proofs below do not use them.

$\forall I$ requires its variable to be absent from every open assumption. $\forall E$ uses capture-free substitution. For $\exists E$, the witness variable is fresh in the goal, the remaining context and the original quantified formula. The kernel uses a conservative, stronger syntactic freshness convention: its distinct-variable conditions forbid the relevant variable names even as bound occurrences. The generated proofs respect that convention.

Equality rules are reflexivity, symmetry, transitivity, congruence for addition and multiplication, and replacement in a formula by capture-free substitution. They are logical equality rules, not arithmetic lemmas. Every replacement in the tables is explicit. Formula formation, substitution and boundedness certificates are syntactic judgments checked separately by the kernel; they do not assert extra facts about numbers.

3 Exactly what a machine and a halting history mean

A configuration is the quadruple $c = (q, L, b, R)$. The state is $q = 0$ (A), $q = 1$ (B), or $q = 2$ (Z, halting); b is the scanned symbol. The least significant bit of L is the nearest cell to the left, and the least significant bit of R is the nearest cell to the right. Higher bits describe cells farther away. A zero tail describes infinitely many blanks. For example, $L = 5$ has binary digits 101, so the three left neighbors, nearest first, contain $1, 0, 1$; all further left cells are blank.

The four instruction terms d_0, d_1, d_2, d_3 stand for A0,A1,B0,B1. Code $4s + 2e + w$ means: write $w \in \{0, 1\}$, move left if $e = 0$ or right if $e = 1$, and enter state $s \in \{0, 1, 2\}$. Validity is the finite formula

$$\mathsf{Valid}(d) := \bigwedge_{j=0}^{3} \bigvee_{k=0}^{11} (d_j = k).$$

The finite conjunctions and disjunctions here and below are literal right-associated first-order formulas, not quantification over sets or lists.

For $c = (q, L, b, R)$ and $c' = (q', L', b', R')$, $\mathsf{Step}_d(c, c')$ is the disjunction of 48 clauses, one for each source state $q_0 \in \{0, 1\}$, read bit $b_0 \in \{0, 1\}$ and instruction code $k \in \{0, \ldots, 11\}$. The clause contains $q = q_0$, $b = b_0$, $d_{2q_0+b_0} = k$, $q' = s$, and $b' = 0 \vee b' = 1$, together with:

$$\begin{aligned} &\text{right:} \quad L' = 2L + w, \quad R = 2R' + b', \\ &\text{left:} \quad R' = 2R + w, \quad L = 2L' + b'. \end{aligned}$$

These equations are the original binary-stack transition relation. They neither add a tape invariant to the machine nor assert that a successor always exists. A proposed run supplies its own successors. State Z has no outgoing clause.

We retain the quotient-and-remainder history code. Fix a duration n, a bound H, and eight coding parameters B_j, C_j $(0 \le j < 4)$. Define

$$\begin{aligned} m_j(i) &:= 1 + (i+1)C_j, \\ \beta_j(i,v) &:= \exists u < H\,[B_j = u \cdot m_j(i) + v \ \wedge\ v < m_j(i)], \\ \mathsf{At}(i;q,L,b,R) &:= \beta_0(i,q) \wedge \big(\beta_1(i,L) \wedge (\beta_2(i,b) \wedge \beta_3(i,R))\big). \end{aligned}$$

A code satisfies $\mathsf{Run}_d(n,H,B,C)$ precisely when all four following formulas hold:

1. $\forall i < n+1\ \exists q,L,b,R < H\ \mathsf{At}(i;q,L,b,R)$;
2. $\forall q,L,b,R < H\,[\mathsf{At}(0;q,L,b,R) \to (q = 0 \wedge (L = 0 \wedge (b = 0 \wedge R = 0)))]$;
3. $\forall i < n\ \forall c,c' < H\,[(\mathsf{At}(i;c) \wedge \mathsf{At}(i+1;c')) \to \mathsf{Step}_d(c,c')]$;
4. $\exists q,L,b,R < H\,[\mathsf{At}(n;q,L,b,R) \wedge q = 2]$.

Here a quantified quadruple means four successive guarded quantifiers, not a new tuple sort; $c,c' < H$ abbreviates eight such quantifiers. Existential blocks are similarly nested. The final halting configuration is explicitly recorded at index n; the transition into it is the instruction from index $n-1$ to n.

$$\mathsf{Halts}_d(n) := \exists H \exists B_0 \exists C_0 \cdots \exists B_3 \exists C_3\ \mathsf{Run}_d(n,H,B,C).$$

"Runs forever" is $\forall n\, \neg\mathsf{Halts}_d(n)$. It denies every proposed coded halting run. No total exponentiation, Chinese remainder theorem, arbitrary sequence-coding theorem, or existence of quotient and remainder is assumed in the soundness proof. This point matters in weak arithmetic. The literal six-step lower bound instead uses an explicit code.

4 A small certificate language for the two loop types

Define a bounded affine pattern by

$$\mathsf{Pat}_{m,p}(x) := \exists u < x+1\,[x = m \cdot u + p].$$

The witness bound is essential: this is a Δ_0 formula. $\mathsf{Pat}_{0,p}(x)$ is equivalent to $x = p$; $\mathsf{Pat}_{1,0}(x)$ holds for every x. $\mathsf{Pat}_{2,1}(x)$ describes an odd stack, and $\mathsf{Pat}_{4,1}(x)$ describes a stack whose two nearest bits are $1,0$. These descriptions are explanations of fixed formulas; exponentiation is not a symbol or an axiom.

A row has eight entries (f,τ,s,a,m,p,r,z) and represents

$$(f = 0 \vee i = \tau) \wedge \big(q = s \wedge (b = a \wedge (\mathsf{Pat}_{m,p}(L) \wedge \mathsf{Pat}_{r,z}(R)))\big).$$

Its first part is a time flag: $f = 0$ makes the row available at every time. A finite halting prefix uses $f = 1$ and a particular time τ. An eight-row chart $I(i;c)$ is the disjunction of its eight row formulas.

An ordinary cycler uses exact stacks. A right translated cycler uses an exact right stack and fixes just the low bits of the left stack that might still be read. The older part of the left stack is left arbitrary. A left translated cycler is symmetric. The important step is to check closure of this *larger* set of configurations. Mere agreement of two simulated local pictures would not suffice.

There are three reusable operations, proved line by line below.

- **Push.** From $\mathsf{Pat}_{m,p}(x)$ and $y = 2x + w$, derive $\mathsf{Pat}_{2m,2p+w}(y)$. The old witness remains bounded by $y+1$ (Lemma 11).
- **Pop.** From $x = 2y + b$ and a pattern fixing the low bit of x, derive both the popped bit and a pattern for y. Binary quotient-and-bit uniqueness is Lemma 16; the even-coefficient and exact cases are Lemmas 20 and 21. No quotient-existence assertion is needed.
- **Inclusion.** $\mathsf{Pat}_{(s+1)k,(s+1)v+r}(x) \to \mathsf{Pat}_{s+1,r}(x)$ (Lemma 22). This proves that the resulting stack pattern belongs to the selected successor row. The exact and unconstrained target cases use Lemmas 17 and 18 instead.

For a finite certificate the checker expands each of its rows against the 48 instruction clauses. A mismatched state, scanned bit or instruction code is excluded by unequal numerals (Lemma 26). A matching instruction uses push, pop and inclusion. The two possible next scanned bits are handled by ordinary $\vee E$. It then uses finitely many $\wedge I$ and $\vee I$ steps to construct the successor row and chart. All fixed numeric identities are expanded into proofs from A1–A8; the delivered traces contain those expansions.

For example, `1RA---_------` has the single row $(0, 0, 0, 0, 1, 0, 0, 0)$: state A, scanned blank, arbitrary left stack, empty right stack. Its instruction pushes 1 on the left and pops the empty right stack. Pop uniqueness forces the next bit and next right stack to be zero. The left stack remains unconstrained. The row therefore implies itself after a step, while excluding state Z. Repeating this row eight times supplies a chart for the common theorem.

This certificate language is not n-gram CPS. It uses no alphabet of position types, no neighborhood closure construction and no CPS saturation. The finite disjunction is simply a direct arithmetic invariant for the loop being certified.

5 The one induction that makes the certificates sound

Let G be the atomic formula $g_0 < g_1$. The following three formulas are the certificate obligations:

$$\begin{aligned}\mathsf{Init} &:= \forall c < H\,[\mathsf{Blank}(c) \to I(0; c)],\\ \mathsf{Close} &:= \forall i\ \forall c, c' < H\,[(I(i; c) \wedge \mathsf{Step}_d(c, c')) \to I(i + 1; c')],\\ \mathsf{Final} &:= \forall c < H\,[(I(n; c) \wedge q = 2) \to G].\end{aligned}$$

Lemma 30 proves, without any additional semantic hypothesis,

$$(\mathsf{Init} \wedge (\mathsf{Close} \wedge (\mathsf{Final} \wedge \mathsf{Run}_d(n, H, B, C)))) \to G.$$

For a nonhalting certificate set G to $0 < 0$ and use O1. For a halting-prefix certificate set G to $n < 7$. State-Z rows then occur only at their permitted finite times, all at most 6. The same theorem handles both uses; it does not require a second history argument.

The induction formula is exactly

$$\varphi(i) := \big(i < n + 1 \to \forall q, L, b, R < H\,[\mathsf{At}(i; q, L, b, R) \to I(i; q, L, b, R)]\big).$$

Everything inside φ is bounded: the quadruple variables are below H, the quotient witnesses in At are below H, and the pattern witnesses are below the corresponding stack plus 1. The row disjunction has exactly eight entries. All machine codes, row entries, n, H, B, C are parameters. Thus φ is Δ_0, and the displayed induction instance is permitted.

The base uses the initial-state clauses of the proposed run. In the successor step, $i + 1 < n + 1$ implies $i < n$. Totality *of this supplied history* gives a configuration at time i; the induction hypothesis puts it in the chart. The run's step clause and certificate closure put the supplied next configuration in the chart. At $i = n$, the final-state clause supplies a configuration with $q = 2$, and the final obligation gives G.

There is no induction on an unbounded assertion such as "every possible tape eventually repeats." Nor do we induct on a statement asserting that a history code exists. Both would need separate boundedness checks. The 352-line table for Lemma 30 records the quantifier introductions, eliminations, witness choices and assumption discharges needed for the permitted induction.

6 Finite coverage and the value

The published BB2 enumeration contains 61 partial tables: 42 certified nonhalting cases and 19 halting frontiers. A halting frontier stops just before a previously undefined instruction is executed. If that instruction is filled by any of codes 8–11, the resulting machine halts one step later. All such times are at most 6.

For the 42 original nonhalting cases the certificates split as 6 ordinary cyclers, 20 right translated cyclers and 16 left translated cyclers. The largest discovery time is 5; budget 4 misses `1RB---_1LB0RB`. The retained-bit coefficients in these certificates are only $0, 1, 2, 4$, and their constant stack parts are at most 1.

To avoid a separate formal theorem about normalization, we check finite coverage directly. Add the left-right reflections of the partial tables and the four-row invariant for tables whose four instructions all return to working states. The resulting 122 arithmetic case conditions cover all $12^4 = 20,736$ tables. A 225-node propositional case tree (183 leaves) proves coverage. A case tree node uses only finite disjunction elimination and earlier case implications. The standalone source contains all these proofs, with no leaf assumption left unproved.

The resulting upper theorem is

$$\forall d_0 \forall d_1 \forall d_2 \forall d_3 \forall n \, [(\mathsf{Valid}(d) \wedge \mathsf{Halts}_d(n)) \to n < 7].$$

For the lower bound the machine `1RB1LB_1LA1RZ` has codes $(7, 5, 1, 11)$ and the following exact trace:

Time	State	L	Scanned bit	R
0	0	0	0	0
1	1	1	0	0
2	0	0	1	1
3	1	0	0	3
4	0	0	0	7
5	1	1	1	3
6	2	3	1	1

The companion proof supplies $C_j = 60$ for all four fields,

$$\begin{aligned}(B_0, B_1, B_2, B_3) = (&457534312199636,\ 14171011907762843,\\ &13837268216790557,\ 14388336956781480),\\ H = {}&14388336956781481.\end{aligned}$$

It proves the quotient-and-remainder equations for every field at every time and constructs the required existential history. These are literal numerical proofs, not an appeal to the totality of sequence coding. Together with the upper bound this yields the exact value 6.

Proof-file interfaces. The final theorem is `bb2-cps-free-exact`. Its first conjunct supplies a valid table halting at time 6; its second conjunct is the universally quantified upper theorem. The database has the same 93 trusted statements as the starting kernel: 16 grammar rules, 18 substitution rules, 15 formula-class rules, 28 logical/equality rules, 15 arithmetic axioms and the bounded-induction rule. Only the last 16 items assert arithmetic or license arithmetic induction. The rest implement syntax and first-order logic. The consolidated database has 9,478 proved statements after merging 10,112 exact duplicate assertions. These counts include the routine numerical proofs for the explicit history; they are not counts of mathematical induction arguments. The new soundness core has one use of the induction rule; the lower-bound arithmetic and the finite coverage do not add induction arguments.

7 The finite certificates

Each row below has the format $(s, a; m, p; r, z)$ for the state, scanned bit and two affine stack patterns. All nonhalting rows have time flag 0. Duplicate padding rows are omitted. The source index is zero-based; the corresponding module is `families/cINDEX.mm`. Unspecified instructions remain arbitrary. These are the 42 original nonhalting entries; their reflected certificates are generated and proved separately.

Index	Partial machine	Type	Certificate rows
1	`ORA---_------`	right shift	$(0,0;1,0;0,0)$
2	`1RA---_------`	right shift	$(0,0;1,0;0,0)$
4	`ORB---_OLA---`	cycle	$(0,0;0,0;0,0)$ $(1,0;0,0;0,0)$
6	`ORB---_1LAOLA`	cycle	$(0,0;0,0;0,0)$ $(1,0;0,0;0,0)$ $(0,0;0,0;0,1)$ $(1,1;0,0;0,0)$
7	`ORB---_1LA1LA`	cycle	$(0,0;0,0;0,0)$ $(1,0;0,0;0,0)$ $(0,0;0,0;0,1)$ $(1,1;0,0;0,0)$
8	`ORB---_1LAORA`	right shift	$(0,0;1,0;0,0)$ $(1,0;2,0;0,0)$ $(0,0;1,0;0,1)$ $(1,1;2,0;0,0)$
9	`ORB---_1LA1RA`	right shift	$(0,0;1,0;0,0)$ $(1,0;2,0;0,0)$ $(0,0;1,0;0,1)$ $(1,1;2,0;0,0)$
10	`ORB---_1LAOLB`	left shift	$(0,0;0,0;0,0)$ $(1,0;0,0;1,0)$ $(0,0;0,0;2,1)$ $(1,1;0,0;1,0)$
11	`ORB---_1LA1LB`	left shift	$(0,0;0,0;0,0)$ $(1,0;0,0;1,0)$ $(0,0;0,0;2,1)$ $(1,1;0,0;1,0)$
12	`ORB---_1LAORB`	right shift	$(0,0;0,0;0,0)$ $(1,0;2,0;0,0)$ $(0,0;1,0;0,1)$ $(1,1;2,0;0,0)$
14	`ORB---_ORA---`	right shift	$(0,0;1,0;0,0)$ $(1,0;2,0;0,0)$
15	`ORB---_1RA---`	right shift	$(0,0;1,0;0,0)$ $(1,0;2,0;0,0)$
16	`ORB---_OLB---`	left shift	$(0,0;0,0;0,0)$ $(1,0;0,0;1,0)$
17	`ORB---_1LB---`	left shift	$(0,0;0,0;0,0)$ $(1,0;0,0;1,0)$
18	`ORB---_ORB---`	right shift	$(0,0;0,0;0,0)$ $(1,0;1,0;0,0)$
19	`ORB---_1RB---`	right shift	$(0,0;0,0;0,0)$ $(1,0;1,0;0,0)$
22	`1RBOLA_OLA---`	left shift	$(0,0;0,0;2,0)$ $(1,0;0,1;1,0)$ $(0,1;0,0;2,0)$
24	`1RBORA_OLA---`	right shift	$(0,0;1,0;0,0)$ $(1,0;2,1;0,0)$ $(0,1;1,0;0,0)$

Index	Partial machine	Type	Certificate rows
25	`1RB1RA_0LA---`	right shift	$(0,0;1,0;0,0)$ $(1,0;2,1;0,0)$ $(0,1;1,0;0,0)$
26	`1RB0LB_0LA---`	left shift	$(0,0;0,0;2,0)$ $(1,0;0,1;1,0)$ $(0,1;0,0;2,0)$ $(1,0;0,0;4,0)$
27	`1RB1LB_0LA---`	left shift	$(0,0;0,0;2,0)$ $(1,0;0,1;1,0)$ $(0,1;0,0;2,0)$ $(1,0;0,0;4,1)$
28	`1RB0RB_0LA---`	cycle	$(0,0;0,0;0,0)$ $(1,0;0,1;0,0)$ $(0,1;0,0;0,0)$ $(1,0;0,0;0,0)$
29	`1RB1RB_0LA---`	cycle	$(0,0;0,0;0,0)$ $(1,0;0,1;0,0)$ $(0,1;0,0;0,0)$
31	`1RB0LA_1LA---`	left shift	$(0,0;0,0;2,0)$ $(1,0;0,1;1,0)$ $(0,1;0,0;2,1)$
33	`1RB0RA_1LA---`	right shift	$(0,0;1,0;0,0)$ $(1,0;2,1;0,0)$ $(0,1;1,0;0,1)$ $(0,1;2,0;0,0)$
34	`1RB1RA_1LA---`	right shift	$(0,0;1,0;0,0)$ $(1,0;2,1;0,0)$ $(0,1;1,0;0,1)$ $(0,1;2,1;0,0)$
39	`1RB---_0RA---`	right shift	$(0,0;1,0;0,0)$ $(1,0;2,1;0,0)$
40	`1RB---_1RA---`	right shift	$(0,0;1,0;0,0)$ $(1,0;2,1;0,0)$
42	`1RB---_0LB0LA`	left shift	$(0,0;0,0;2,0)$ $(1,0;0,1;1,0)$ $(1,1;0,0;2,0)$
44	`1RB---_0LB0RA`	right shift	$(0,0;1,0;0,0)$ $(1,0;2,1;0,0)$ $(1,1;1,0;0,0)$
45	`1RB---_0LB1RA`	right shift	$(0,0;1,0;0,0)$ $(1,0;2,1;0,0)$ $(1,1;1,0;0,0)$
46	`1RB---_0LB0LB`	left shift	$(0,0;0,0;0,0)$ $(1,0;0,1;0,0)$ $(1,1;0,0;0,0)$ $(1,0;0,0;1,0)$
47	`1RB---_0LB1LB`	left shift	$(0,0;0,0;0,0)$ $(1,0;0,1;0,0)$ $(1,1;0,0;0,0)$ $(1,0;0,0;1,0)$
48	`1RB---_0LB0RB`	left shift	$(0,0;0,0;0,0)$ $(1,0;0,1;0,0)$ $(1,1;0,0;0,0)$ $(1,0;0,0;1,0)$
49	`1RB---_0LB1RB`	cycle	$(0,0;0,0;0,0)$ $(1,0;0,1;0,0)$ $(1,1;0,0;0,0)$

Index	Partial machine	Type	Certificate rows
51	`1RB---_1LB0LA`	left shift	$(0,0;0,0;2,0)$ $(1,0;0,1;1,0)$ $(1,1;0,0;2,1)$
55	`1RB---_1LB0LB`	left shift	$(0,0;0,0;0,0)$ $(1,0;0,1;0,0)$ $(1,1;0,0;0,1)$ $(1,0;0,0;1,0)$
56	`1RB---_1LB1LB`	left shift	$(0,0;0,0;0,0)$ $(1,0;0,1;0,0)$ $(1,1;0,0;0,1)$ $(1,0;0,0;1,0)$
57	`1RB---_1LB0RB`	left shift	$(0,0;0,0;0,0)$ $(1,0;0,1;0,0)$ $(1,1;0,0;0,1)$ $(1,1;0,0;0,0)$ $(1,0;0,0;1,0)$
58	`1RB---_1LB1RB`	right shift	$(0,0;0,0;0,0)$ $(1,0;2,1;0,0)$ $(1,1;1,0;0,1)$ $(1,1;2,1;0,0)$
59	`1RB---_0RB---`	right shift	$(0,0;0,0;0,0)$ $(1,0;1,0;0,0)$
60	`1RB---_1RB---`	right shift	$(0,0;0,0;0,0)$ $(1,0;1,0;0,0)$

7.1 Halting frontiers

Each time is the number of already executed working-state instructions. Replacing the next undefined instruction by a halting instruction adds one step.

Index	Partial machine	Frontier time
0	`------_------`	0
3	`0RB---_------`	1
5	`0RB---_1LA---`	3
13	`0RB---_1LA1RB`	5
20	`1RB---_------`	1
21	`1RB---_0LA---`	2
23	`1RB1LA_0LA---`	4
30	`1RB---_1LA---`	2
32	`1RB1LA_1LA---`	4
35	`1RB0LB_1LA---`	5
36	`1RB1LB_1LA---`	5
37	`1RB0RB_1LA---`	3
38	`1RB1RB_1LA---`	3
41	`1RB---_0LB---`	2
43	`1RB---_0LB1LA`	5
50	`1RB---_1LB---`	2
52	`1RB---_1LB1LA`	5
53	`1RB---_1LB0RA`	3
54	`1RB---_1LB1RA`	3

8 Exact natural-deduction proofs of the shared core

Every table below is generated from the same proof construction that emits the checked Metamath statements. No call to a ring tactic, computer search or external arithmetic oracle is a proof rule. A reference to an earlier lemma imports that closed theorem; the ensuing universal eliminations are separate lines.

For compactness, F_j is a *textual formula abbreviation*, local to its table, not a new predicate or propositional constant. Its definition preserves all free and bound variable occurrences. In $\forall x\, F_j$, first expand F_j literally, then apply the quantifier; the variable may occur inside its expansion. Definitions are acyclic. The variable key gives the exact correspondence with the machine-readable trace. The abbreviation Pat always expands to its explicitly bounded definition above. Assumption labels are likewise only abbreviations for the displayed formulas. Consequently every sequent expands to first-order logic in the original arithmetic language.

8.1 zero left

Statement. $\forall x_1\,(0 + x_1) = x_1$.

Variable key. `zLeftX` $= x_1$.

Line	Open assumptions	Formula	Rule and premises
1	$\varnothing$	$(0 + x_1) = (x_1 + 0)$	Axiom A2
2	$\varnothing$	$(x_1 + 0) = x_1$	Axiom A3
3	$\varnothing$	$(0 + x_1) = x_1$	= trans 1,2
4	$\varnothing$	$\forall x_1\,(0 + x_1) = x_1$	$\forall I$ 3

8.2 below successor

Statement. $\forall x_1\, x_1 < (x_1 + 1)$.

Variable key. `e1` $= x_2$, `succX` $= x_1$.

Line	Open assumptions	Formula	Rule and premises
1	$\varnothing$	$(x_1 < (x_1 + 1) \leftrightarrow \exists x_2\, (x_1 + 1) = ((x_1 + x_2) + 1))$	Axiom O7
2	$\varnothing$	$(\exists x_2\, (x_1 + 1) = ((x_1 + x_2) + 1) \rightarrow x_1 < (x_1 + 1))$	$\leftrightarrow E_2$ 1
3	$\varnothing$	$(x_1 + 0) = x_1$	Axiom A3
4	$\varnothing$	$1 = 1$	$= I$
5	$\varnothing$	$((x_1 + 0) + 1) = (x_1 + 1)$	+cong 3,4
6	$\varnothing$	$(x_1 + 1) = ((x_1 + 0) + 1)$	= sym 5
7	$\varnothing$	$\exists x_2\, (x_1 + 1) = ((x_1 + x_2) + 1)$	$\exists I$ 6
8	$\varnothing$	$x_1 < (x_1 + 1)$	$\rightarrow E$ 2,7
9	$\varnothing$	$\forall x_1\, x_1 < (x_1 + 1)$	$\forall I$ 8

8.3 successor positive

Statement. $\forall x_1\, 0 < (x_1 + 1)$.

Variable key. `e1` $= x_2$, `posX` $= x_1$, `zLeftX` $= x_3$.

Line	Open assumptions	Formula	Rule and premises
1	$\varnothing$	$(0 < (x_1 + 1) \leftrightarrow \exists x_2\, (x_1 + 1) = ((0 + x_2) + 1))$	Axiom O7
2	$\varnothing$	$(\exists x_2\, (x_1 + 1) = ((0 + x_2) + 1) \rightarrow 0 < (x_1 + 1))$	$\leftrightarrow E_2$ 1
3	$\varnothing$	$\forall x_3\, (0 + x_3) = x_3$	Lemma 1
4	$\varnothing$	$\forall x_3\, (0 + x_3) = x_3$	Closed-Weak 3
5	$\varnothing$	$(0 + x_1) = x_1$	$\forall E$ 4
6	$\varnothing$	$1 = 1$	$= I$
7	$\varnothing$	$((0 + x_1) + 1) = (x_1 + 1)$	+cong 5,6
8	$\varnothing$	$(x_1 + 1) = ((0 + x_1) + 1)$	= sym 7
9	$\varnothing$	$\exists x_2\, (x_1 + 1) = ((0 + x_2) + 1)$	$\exists I$ 8
10	$\varnothing$	$0 < (x_1 + 1)$	$\rightarrow E$ 2,9
11	$\varnothing$	$\forall x_1\, 0 < (x_1 + 1)$	$\forall I$ 10

8.4 plus cancel

Statement. $\forall x_1\,\forall x_3\,\forall x_2\,((x_1 + x_2) = (x_3 + x_2) \to x_1 = x_3)$.

Variable key. $\texttt{canX} = x_1$, $\texttt{canY} = x_3$, $\texttt{canZ} = x_2$.

Assumption key.

a_1	$(x_1 + x_2) = (x_3 + x_2)$
a_2	$x_1 < x_3$
a_3	$(x_1 = x_3 \vee x_3 < x_1)$
a_4	$x_1 = x_3$
a_5	$x_3 < x_1$

Line	Open assumptions	Formula	Rule and premises
1	a_1	$(x_1 < x_3 \vee (x_1 = x_3 \vee x_3 < x_1))$	Axiom O3
2	a_1	$(x_1 + x_2) = (x_3 + x_2)$	Assume
3	a_1, a_2	$(x_1 + x_2) = (x_3 + x_2)$	Weak 2
4	a_1, a_2	$(x_3 + x_2) = (x_1 + x_2)$	$=$ sym 3
5	a_1, a_2	$(x_1 < x_3 \to (x_1 + x_2) < (x_3 + x_2))$	Axiom O5
6	a_1, a_2	$x_1 < x_3$	Assume
7	a_1, a_2	$(x_1 + x_2) < (x_3 + x_2)$	$\to E$ 5,6
8	a_1, a_2	$(x_1 + x_2) < (x_1 + x_2)$	$= E$ 4,7
9	a_1, a_2	$\neg(x_1 + x_2) < (x_1 + x_2)$	Axiom O1
10	a_1, a_2	$x_1 = x_3$	$\bot E$ 8,9
11	a_1, a_3	$(x_1 = x_3 \vee x_3 < x_1)$	Assume
12	a_1, a_3, a_4	$x_1 = x_3$	Assume
13	a_1, a_3	$(x_1 + x_2) = (x_3 + x_2)$	Weak 2
14	a_1, a_3, a_5	$(x_1 + x_2) = (x_3 + x_2)$	Weak 13
15	a_1, a_3, a_5	$(x_3 < x_1 \to (x_3 + x_2) < (x_1 + x_2))$	Axiom O5
16	a_1, a_3, a_5	$x_3 < x_1$	Assume
17	a_1, a_3, a_5	$(x_3 + x_2) < (x_1 + x_2)$	$\to E$ 15,16
18	a_1, a_3, a_5	$(x_3 + x_2) < (x_3 + x_2)$	$= E$ 14,17
19	a_1, a_3, a_5	$\neg(x_3 + x_2) < (x_3 + x_2)$	Axiom O1
20	a_1, a_3, a_5	$x_1 = x_3$	$\bot E$ 18,19
21	a_1, a_3	$x_1 = x_3$	$\vee E$ 11,12,20
22	a_1	$x_1 = x_3$	$\vee E$ 1,10,21
23	$\varnothing$	$((x_1 + x_2) = (x_3 + x_2) \to x_1 = x_3)$	$\to I$ 22
24	$\varnothing$	$\forall x_2\,((x_1 + x_2) = (x_3 + x_2) \to x_1 = x_3)$	$\forall I$ 23
25	$\varnothing$	$\forall x_3\,\forall x_2\,((x_1 + x_2) = (x_3 + x_2) \to x_1 = x_3)$	$\forall I$ 24
26	$\varnothing$	$\forall x_1\,\forall x_3\,\forall x_2\,((x_1 + x_2) = (x_3 + x_2) \to x_1 = x_3)$	$\forall I$ 25

8.5 rotate sum

Statement. $\forall x_1\,\forall x_3\,\forall x_2\,((x_1 + x_3) + x_2) = ((x_1 + x_2) + x_3)$.

Variable key. `rotX` $= x_1$, `rotY` $= x_3$, `rotZ` $= x_2$.

Line	Open assumptions	Formula	Rule and premises
1	$\varnothing$	$((x_1 + x_2) + x_3) = (x_1 + (x_2 + x_3))$	Axiom A1
2	$\varnothing$	$(x_1 + (x_2 + x_3)) = ((x_1 + x_2) + x_3)$	= sym 1
3	$\varnothing$	$(x_3 + x_2) = (x_2 + x_3)$	Axiom A2
4	$\varnothing$	$((x_1 + x_3) + x_2) = (x_1 + (x_3 + x_2))$	Axiom A1
5	$\varnothing$	$((x_1 + x_3) + x_2) = ((x_1 + x_3) + x_2)$	$= I$
6	$\varnothing$	$((x_1 + x_3) + x_2) = (x_1 + (x_3 + x_2))$	$= E$ 4,5
7	$\varnothing$	$((x_1 + x_3) + x_2) = (x_1 + (x_2 + x_3))$	$= E$ 3,6
8	$\varnothing$	$((x_1 + x_3) + x_2) = ((x_1 + x_2) + x_3)$	$= E$ 2,7
9	$\varnothing$	$\forall x_2\,((x_1 + x_3) + x_2) = ((x_1 + x_2) + x_3)$	$\forall I$ 8
10	$\varnothing$	$\forall x_3\,\forall x_2\,((x_1 + x_3) + x_2) = ((x_1 + x_2) + x_3)$	$\forall I$ 9
11	$\varnothing$	$\forall x_1\,\forall x_3\,\forall x_2\,((x_1 + x_3) + x_2) = ((x_1 + x_2) + x_3)$	$\forall I$ 10

8.6 double sum

Statement. $\forall x_1\,(2 \cdot x_1) = (x_1 + x_1)$.

Variable key. `doubleX` $= x_1$.

Line	Open assumptions	Formula	Rule and premises
1	$\varnothing$	$(x_1 \cdot 1) = x_1$	Axiom A6
2	$\varnothing$	$(x_1 \cdot 2) = ((x_1 \cdot 1) + (x_1 \cdot 1))$	Axiom A8
3	$\varnothing$	$(2 \cdot x_1) = (x_1 \cdot 2)$	Axiom A5
4	$\varnothing$	$(2 \cdot x_1) = (2 \cdot x_1)$	$= I$
5	$\varnothing$	$(2 \cdot x_1) = (x_1 \cdot 2)$	$= E$ 3,4
6	$\varnothing$	$(2 \cdot x_1) = ((x_1 \cdot 1) + (x_1 \cdot 1))$	$= E$ 2,5
7	$\varnothing$	$(2 \cdot x_1) = (x_1 + (x_1 \cdot 1))$	$= E$ 1,6
8	$\varnothing$	$(2 \cdot x_1) = (x_1 + x_1)$	$= E$ 1,7
9	$\varnothing$	$\forall x_1\,(2 \cdot x_1) = (x_1 + x_1)$	$\forall I$ 8

8.7 factor identity

Statement. $\forall x_1\,\forall x_2\,\forall x_3\,\forall x_4\,\forall x_5\,F_1$.

Variable key. $\texttt{facA} = x_1$, $\texttt{facB} = x_2$, $\texttt{facC} = x_3$, $\texttt{facD} = x_4$, $\texttt{facE} = x_5$.

Textual formula abbreviations. Expand these from top to bottom; each right-hand side refers only to earlier abbreviations.

F_1 $(((x_1 \cdot x_2) \cdot x_3) + ((x_1 \cdot x_4) + x_5)) = ((x_1 \cdot ((x_2 \cdot x_3) + x_4)) + x_5)$

F_2 $(((x_1 \cdot (x_2 \cdot x_3)) + (x_1 \cdot x_4)) + x_5) = ((x_1 \cdot (x_2 \cdot x_3)) + ((x_1 \cdot x_4) + x_5))$

F_3 $((x_1 \cdot (x_2 \cdot x_3)) + ((x_1 \cdot x_4) + x_5)) = (((x_1 \cdot (x_2 \cdot x_3)) + (x_1 \cdot x_4)) + x_5)$

F_4 $(((x_1 \cdot x_2) \cdot x_3) + ((x_1 \cdot x_4) + x_5)) = (((x_1 \cdot x_2) \cdot x_3) + ((x_1 \cdot x_4) + x_5))$

F_5 $(((x_1 \cdot x_2) \cdot x_3) + ((x_1 \cdot x_4) + x_5)) = ((x_1 \cdot (x_2 \cdot x_3)) + ((x_1 \cdot x_4) + x_5))$

F_6 $(((x_1 \cdot x_2) \cdot x_3) + ((x_1 \cdot x_4) + x_5)) = (((x_1 \cdot (x_2 \cdot x_3)) + (x_1 \cdot x_4)) + x_5)$

Line	Open assumptions	Formula	Rule and premises
1	$\varnothing$	$(x_1 \cdot ((x_2 \cdot x_3) + x_4)) = ((x_1 \cdot (x_2 \cdot x_3)) + (x_1 \cdot x_4))$	Axiom A8
2	$\varnothing$	$((x_1 \cdot (x_2 \cdot x_3)) + (x_1 \cdot x_4)) = (x_1 \cdot ((x_2 \cdot x_3) + x_4))$	= sym 1
3	$\varnothing$	F_2	Axiom A1
4	$\varnothing$	F_3	= sym 3
5	$\varnothing$	$((x_1 \cdot x_2) \cdot x_3) = (x_1 \cdot (x_2 \cdot x_3))$	Axiom A4
6	$\varnothing$	F_4	$= I$
7	$\varnothing$	F_5	$= E$ 5,6
8	$\varnothing$	F_6	$= E$ 4,7
9	$\varnothing$	F_1	$= E$ 2,8
10	$\varnothing$	$\forall x_5\,F_1$	$\forall I$ 9
11	$\varnothing$	$\forall x_4\,\forall x_5\,F_1$	$\forall I$ 10
12	$\varnothing$	$\forall x_3\,\forall x_4\,\forall x_5\,F_1$	$\forall I$ 11
13	$\varnothing$	$\forall x_2\,\forall x_3\,\forall x_4\,\forall x_5\,F_1$	$\forall I$ 12
14	$\varnothing$	$\forall x_1\,\forall x_2\,\forall x_3\,\forall x_4\,\forall x_5\,F_1$	$\forall I$ 13

8.8 coefficient gap

Statement. F_1.

Variable key. `coefP` $= x_3$, `coefS` $= x_2$, `coefU` $= x_1$.

Textual formula abbreviations. Expand these from top to bottom; each right-hand side refers only to earlier abbreviations.

F_1 $\quad \forall x_2\, \forall x_1\, \forall x_3\, (((x_2 + 1) \cdot x_1) + x_3) = (x_1 + ((x_2 \cdot x_1) + x_3))$

Line	Open assumptions	Formula	Rule and premises
1	$\varnothing$	$((x_1 + (x_2 \cdot x_1)) + x_3) = (x_1 + ((x_2 \cdot x_1) + x_3))$	Axiom A1
2	$\varnothing$	$((x_2 \cdot x_1) + x_1) = (x_1 + (x_2 \cdot x_1))$	Axiom A2
3	$\varnothing$	$(x_1 \cdot x_2) = (x_2 \cdot x_1)$	Axiom A5
4	$\varnothing$	$(x_1 \cdot 1) = x_1$	Axiom A6
5	$\varnothing$	$(x_1 \cdot (x_2 + 1)) = ((x_1 \cdot x_2) + (x_1 \cdot 1))$	Axiom A8
6	$\varnothing$	$((x_2 + 1) \cdot x_1) = (x_1 \cdot (x_2 + 1))$	Axiom A5
7	$\varnothing$	$(((x_2 + 1) \cdot x_1) + x_3) = (((x_2 + 1) \cdot x_1) + x_3)$	$= I$
8	$\varnothing$	$(((x_2 + 1) \cdot x_1) + x_3) = ((x_1 \cdot (x_2 + 1)) + x_3)$	$= E$ 6,7
9	$\varnothing$	$(((x_2 + 1) \cdot x_1) + x_3) = (((x_1 \cdot x_2) + (x_1 \cdot 1)) + x_3)$	$= E$ 5,8
10	$\varnothing$	$(((x_2 + 1) \cdot x_1) + x_3) = (((x_1 \cdot x_2) + x_1) + x_3)$	$= E$ 4,9
11	$\varnothing$	$(((x_2 + 1) \cdot x_1) + x_3) = (((x_2 \cdot x_1) + x_1) + x_3)$	$= E$ 3,10
12	$\varnothing$	$(((x_2 + 1) \cdot x_1) + x_3) = ((x_1 + (x_2 \cdot x_1)) + x_3)$	$= E$ 2,11
13	$\varnothing$	$(((x_2 + 1) \cdot x_1) + x_3) = (x_1 + ((x_2 \cdot x_1) + x_3))$	$= E$ 1,12
14	$\varnothing$	$\forall x_3\, (((x_2 + 1) \cdot x_1) + x_3) = (x_1 + ((x_2 \cdot x_1) + x_3))$	$\forall I$ 13
15	$\varnothing$	$\forall x_1\, \forall x_3\, (((x_2 + 1) \cdot x_1) + x_3) = (x_1 + ((x_2 \cdot x_1) + x_3))$	$\forall I$ 14
16	$\varnothing$	F_1	$\forall I$ 15

8.9 coefficient bound

Statement. $\forall x_1\, F_1$.

Variable key. $\texttt{bdP} = x_4$, $\texttt{bdS} = x_2$, $\texttt{bdU} = x_3$, $\texttt{bdX} = x_1$, $\texttt{coefP} = x_8$, $\texttt{coefS} = x_6$, $\texttt{coefU} = x_7$, $\texttt{e1} = x_5$.

Textual formula abbreviations. Expand these from top to bottom; each right-hand side refers only to earlier abbreviations.

F_1 $\quad \forall x_2\, \forall x_3\, \forall x_4\, (x_1 = (((x_2 + 1) \cdot x_3) + x_4) \to x_3 < (x_1 + 1))$

F_2 $\quad \forall x_6\, \forall x_7\, \forall x_8\, (((x_6 + 1) \cdot x_7) + x_8) = (x_7 + ((x_6 \cdot x_7) + x_8))$

Assumption key.

a_1 $\quad x_1 = (((x_2 + 1) \cdot x_3) + x_4)$

Line	Open assumptions	Formula	Rule and premises
1	a_1	$(x_3 < (x_1 + 1) \leftrightarrow \exists x_5\, (x_1 + 1) = ((x_3 + x_5) + 1))$	Axiom O7
2	a_1	$(\exists x_5\, (x_1 + 1) = ((x_3 + x_5) + 1) \to x_3 < (x_1 + 1))$	$\leftrightarrow E_2$ 1
3	a_1	$x_1 = (((x_2 + 1) \cdot x_3) + x_4)$	Assume
4	$\varnothing$	F_2	Lemma 8
5	a_1	F_2	Closed-Weak 4
6	a_1	$\forall x_7\, \forall x_8\, (((x_2 + 1) \cdot x_7) + x_8) = (x_7 + ((x_2 \cdot x_7) + x_8))$	$\forall E$ 5
7	a_1	$\forall x_8\, (((x_2 + 1) \cdot x_3) + x_8) = (x_3 + ((x_2 \cdot x_3) + x_8))$	$\forall E$ 6
8	a_1	$(((x_2 + 1) \cdot x_3) + x_4) = (x_3 + ((x_2 \cdot x_3) + x_4))$	$\forall E$ 7
9	a_1	$x_1 = (x_3 + ((x_2 \cdot x_3) + x_4))$	$=$ trans 3,8
10	a_1	$1 = 1$	$= I$
11	a_1	$(x_1 + 1) = ((x_3 + ((x_2 \cdot x_3) + x_4)) + 1)$	+cong 9,10
12	a_1	$\exists x_5\, (x_1 + 1) = ((x_3 + x_5) + 1)$	$\exists I$ 11
13	a_1	$x_3 < (x_1 + 1)$	$\to E$ 2,12
14	$\varnothing$	$(x_1 = (((x_2 + 1) \cdot x_3) + x_4) \to x_3 < (x_1 + 1))$	$\to I$ 13
15	$\varnothing$	$\forall x_4\, (x_1 = (((x_2 + 1) \cdot x_3) + x_4) \to x_3 < (x_1 + 1))$	$\forall I$ 14
16	$\varnothing$	$\forall x_3\, \forall x_4\, (x_1 = (((x_2 + 1) \cdot x_3) + x_4) \to x_3 < (x_1 + 1))$	$\forall I$ 15
17	$\varnothing$	F_1	$\forall I$ 16
18	$\varnothing$	$\forall x_1\, F_1$	$\forall I$ 17

8.10 bound gap

Statement. $\forall x_1\,\forall x_2\,(x_1 < (x_2 + 1) \to \exists x_8\,x_2 = (x_1 + x_8))$.

Variable key. $\texttt{canX} = x_4$, $\texttt{canY} = x_5$, $\texttt{canZ} = x_6$, $\texttt{e1} = x_3$, $\texttt{e2} = x_7$, $\texttt{gapU} = x_1$, $\texttt{gapX} = x_2$, $\texttt{j} = x_8$.

Assumption key.

a_1	$x_1 < (x_2 + 1)$
a_2	$(x_2 + 1) = ((x_1 + x_7) + 1)$

Line	Open assumptions	Formula	Rule and premises
1	a_1	$(x_1 < (x_2 + 1) \leftrightarrow \exists x_3\,(x_2 + 1) = ((x_1 + x_3) + 1))$	Axiom O7
2	a_1	$(x_1 < (x_2 + 1) \to \exists x_3\,(x_2 + 1) = ((x_1 + x_3) + 1))$	$\leftrightarrow E_1$ 1
3	a_1	$x_1 < (x_2 + 1)$	Assume
4	a_1	$\exists x_3\,(x_2 + 1) = ((x_1 + x_3) + 1)$	$\to E$ 2,3
5	$\varnothing$	$\forall x_4\,\forall x_5\,\forall x_6\,((x_4 + x_6) = (x_5 + x_6) \to x_4 = x_5)$	Lemma 4
6	a_1, a_2	$\forall x_4\,\forall x_5\,\forall x_6\,((x_4 + x_6) = (x_5 + x_6) \to x_4 = x_5)$	Closed-Weak 5
7	a_1, a_2	$\forall x_5\,\forall x_6\,((x_2 + x_6) = (x_5 + x_6) \to x_2 = x_5)$	$\forall E$ 6
8	a_1, a_2	$\forall x_6\,((x_2 + x_6) = ((x_1 + x_7) + x_6) \to x_2 = (x_1 + x_7))$	$\forall E$ 7
9	a_1, a_2	$((x_2 + 1) = ((x_1 + x_7) + 1) \to x_2 = (x_1 + x_7))$	$\forall E$ 8
10	a_1, a_2	$(x_2 + 1) = ((x_1 + x_7) + 1)$	Assume
11	a_1, a_2	$x_2 = (x_1 + x_7)$	$\to E$ 9,10
12	a_1, a_2	$\exists x_8\,x_2 = (x_1 + x_8)$	$\exists I$ 11
13	a_1	$\exists x_8\,x_2 = (x_1 + x_8)$	$\exists E$ 4,12
14	$\varnothing$	$(x_1 < (x_2 + 1) \to \exists x_8\,x_2 = (x_1 + x_8))$	$\to I$ 13
15	$\varnothing$	$\forall x_2\,(x_1 < (x_2 + 1) \to \exists x_8\,x_2 = (x_1 + x_8))$	$\forall I$ 14
16	$\varnothing$	$\forall x_1\,\forall x_2\,(x_1 < (x_2 + 1) \to \exists x_8\,x_2 = (x_1 + x_8))$	$\forall I$ 15

8.11 push bound

Statement. $\forall x_4\, \forall x_5\, F_1$.

Variable key. $\texttt{doubleX} = x_{10}$, $\texttt{e1} = x_9$, $\texttt{e2} = x_8$, $\texttt{gapU} = x_1$, $\texttt{gapX} = x_2$, $\texttt{j} = x_3$, $\texttt{trU} = x_4$, $\texttt{trW} = x_7$, $\texttt{trX} = x_5$, $\texttt{trY} = x_6$.

Textual formula abbreviations. Expand these from top to bottom; each right-hand side refers only to earlier abbreviations.

F_1 $\quad \forall x_6\, \forall x_7\, ((x_4 < (x_5 + 1) \wedge x_6 = ((2 \cdot x_5) + x_7)) \to x_4 < (x_6 + 1))$

Assumption key.

a_1 $\quad (x_4 < (x_5 + 1) \wedge x_6 = ((2 \cdot x_5) + x_7))$

a_2 $\quad x_5 = (x_4 + x_8)$

Line	Open assumptions	Formula	Rule and premises
1	$\varnothing$	$\forall x_1\, \forall x_2\, (x_1 < (x_2 + 1) \to \exists x_3\, x_2 = (x_1 + x_3))$	Lemma 10
2	a_1	$\forall x_1\, \forall x_2\, (x_1 < (x_2 + 1) \to \exists x_3\, x_2 = (x_1 + x_3))$	Closed-Weak 1
3	a_1	$\forall x_2\, (x_4 < (x_2 + 1) \to \exists x_3\, x_2 = (x_4 + x_3))$	$\forall E$ 2
4	a_1	$(x_4 < (x_5 + 1) \to \exists x_3\, x_5 = (x_4 + x_3))$	$\forall E$ 3
5	a_1	$(x_4 < (x_5 + 1) \wedge x_6 = ((2 \cdot x_5) + x_7))$	Assume
6	a_1	$x_4 < (x_5 + 1)$	$\wedge E_1$ 5
7	a_1	$\exists x_3\, x_5 = (x_4 + x_3)$	$\to E$ 4,6
8	a_1, a_2	$(x_4 < (x_6 + 1) \leftrightarrow \exists x_9\, (x_6 + 1) = ((x_4 + x_9) + 1))$	Axiom O7
9	a_1, a_2	$(\exists x_9\, (x_6 + 1) = ((x_4 + x_9) + 1) \to x_4 < (x_6 + 1))$	$\leftrightarrow E_2$ 8
10	a_1, a_2	$((x_4 + x_8) + (x_5 + x_7)) = (x_4 + (x_8 + (x_5 + x_7)))$	Axiom A1
11	a_1, a_2	$x_5 = (x_4 + x_8)$	Assume
12	a_1, a_2	$((x_5 + x_5) + x_7) = (x_5 + (x_5 + x_7))$	Axiom A1
13	$\varnothing$	$\forall x_{10}\, (2 \cdot x_{10}) = (x_{10} + x_{10})$	Lemma 6
14	a_1, a_2	$\forall x_{10}\, (2 \cdot x_{10}) = (x_{10} + x_{10})$	Closed-Weak 13
15	a_1, a_2	$(2 \cdot x_5) = (x_5 + x_5)$	$\forall E$ 14
16	a_1	$x_6 = ((2 \cdot x_5) + x_7)$	$\wedge E_2$ 5
17	a_1, a_2	$x_6 = ((2 \cdot x_5) + x_7)$	Weak 16
18	a_1, a_2	$(x_6 + 1) = (x_6 + 1)$	$= I$
19	a_1, a_2	$(x_6 + 1) = (((2 \cdot x_5) + x_7) + 1)$	$= E$ 17,18
20	a_1, a_2	$(x_6 + 1) = (((x_5 + x_5) + x_7) + 1)$	$= E$ 15,19
21	a_1, a_2	$(x_6 + 1) = ((x_5 + (x_5 + x_7)) + 1)$	$= E$ 12,20
22	a_1, a_2	$(x_6 + 1) = (((x_4 + x_8) + (x_5 + x_7)) + 1)$	$= E$ 11,21
23	a_1, a_2	$(x_6 + 1) = ((x_4 + (x_8 + (x_5 + x_7))) + 1)$	$= E$ 10,22
24	a_1, a_2	$\exists x_9\, (x_6 + 1) = ((x_4 + x_9) + 1)$	$\exists I$ 23
25	a_1, a_2	$x_4 < (x_6 + 1)$	$\to E$ 9,24
26	a_1	$x_4 < (x_6 + 1)$	$\exists E$ 7,25
27	$\varnothing$	$((x_4 < (x_5 + 1) \wedge x_6 = ((2 \cdot x_5) + x_7)) \to x_4 < (x_6 + 1))$	$\to I$ 26
28	$\varnothing$	$\forall x_7\, ((x_4 < (x_5 + 1) \wedge x_6 = ((2 \cdot x_5) + x_7)) \to x_4 < (x_6 + 1))$	$\forall I$ 27
29	$\varnothing$	F_1	$\forall I$ 28
30	$\varnothing$	$\forall x_5\, F_1$	$\forall I$ 29
31	$\varnothing$	$\forall x_4\, \forall x_5\, F_1$	$\forall I$ 30

8.12 cancel left

Statement. $\forall x_4\,\forall x_5\,\forall x_6\,((x_4 + x_5) = (x_4 + x_6) \to x_5 = x_6)$.

Variable key. $\texttt{canX} = x_1$, $\texttt{canY} = x_2$, $\texttt{canZ} = x_3$, $\texttt{lcX} = x_4$, $\texttt{lcY} = x_5$, $\texttt{lcZ} = x_6$.

Assumption key.

a_1 $(x_4 + x_5) = (x_4 + x_6)$

Line	Open assumptions	Formula	Rule and premises
1	$\varnothing$	$\forall x_1\,\forall x_2\,\forall x_3\,((x_1 + x_3) = (x_2 + x_3) \to x_1 = x_2)$	Lemma 4
2	a_1	$\forall x_1\,\forall x_2\,\forall x_3\,((x_1 + x_3) = (x_2 + x_3) \to x_1 = x_2)$	Closed-Weak 1
3	a_1	$\forall x_2\,\forall x_3\,((x_5 + x_3) = (x_2 + x_3) \to x_5 = x_2)$	$\forall E$ 2
4	a_1	$\forall x_3\,((x_5 + x_3) = (x_6 + x_3) \to x_5 = x_6)$	$\forall E$ 3
5	a_1	$((x_5 + x_4) = (x_6 + x_4) \to x_5 = x_6)$	$\forall E$ 4
6	a_1	$(x_5 + x_4) = (x_4 + x_5)$	Axiom A2
7	a_1	$(x_4 + x_5) = (x_4 + x_6)$	Assume
8	a_1	$(x_5 + x_4) = (x_4 + x_6)$	= trans 6,7
9	a_1	$(x_4 + x_6) = (x_6 + x_4)$	Axiom A2
10	a_1	$(x_5 + x_4) = (x_6 + x_4)$	= trans 8,9
11	a_1	$x_5 = x_6$	$\to E$ 5,10
12	$\varnothing$	$((x_4 + x_5) = (x_4 + x_6) \to x_5 = x_6)$	$\to I$ 11
13	$\varnothing$	$\forall x_6\,((x_4 + x_5) = (x_4 + x_6) \to x_5 = x_6)$	$\forall I$ 12
14	$\varnothing$	$\forall x_5\,\forall x_6\,((x_4 + x_5) = (x_4 + x_6) \to x_5 = x_6)$	$\forall I$ 13
15	$\varnothing$	$\forall x_4\,\forall x_5\,\forall x_6\,((x_4 + x_5) = (x_4 + x_6) \to x_5 = x_6)$	$\forall I$ 14

8.13 no two below

Statement. $\forall x_1\, \forall x_2\, (x_1 < 2 \to \neg x_1 = (x_2 + 2))$.

Variable key. `e1` $= x_3$, `e2` $= x_4$, `ntX` $= x_1$, `ntY` $= x_2$, `rotX` $= x_5$, `rotY` $= x_6$, `rotZ` $= x_7$.

Assumption key.

a_1	$x_1 < 2$
a_2	$x_1 = (x_2 + 2)$
a_3	$2 = ((x_1 + x_4) + 1)$

Line	Open assumptions	Formula	Rule and premises
1	a_1, a_2	$(x_1 < 2 \leftrightarrow \exists x_3\, 2 = ((x_1 + x_3) + 1))$	Axiom O7
2	a_1, a_2	$(x_1 < 2 \to \exists x_3\, 2 = ((x_1 + x_3) + 1))$	$\leftrightarrow E_1$ 1
3	a_1	$x_1 < 2$	Assume
4	a_1, a_2	$x_1 < 2$	Weak 3
5	a_1, a_2	$\exists x_3\, 2 = ((x_1 + x_3) + 1)$	$\to E$ 2,4
6	a_1, a_2, a_3	$(2 < 2 \leftrightarrow \exists x_3\, 2 = ((2 + x_3) + 1))$	Axiom O7
7	a_1, a_2, a_3	$(\exists x_3\, 2 = ((2 + x_3) + 1) \to 2 < 2)$	$\leftrightarrow E_2$ 6
8	a_1, a_2, a_3	$((x_2 + x_4) + 2) = (2 + (x_2 + x_4))$	Axiom A2
9	$\varnothing$	$\forall x_5\, \forall x_6\, \forall x_7\, ((x_5 + x_6) + x_7) = ((x_5 + x_7) + x_6)$	Lemma 5
10	a_1, a_2, a_3	$\forall x_5\, \forall x_6\, \forall x_7\, ((x_5 + x_6) + x_7) = ((x_5 + x_7) + x_6)$	Closed-Weak 9
11	a_1, a_2, a_3	$\forall x_6\, \forall x_7\, ((x_2 + x_6) + x_7) = ((x_2 + x_7) + x_6)$	$\forall E$ 10
12	a_1, a_2, a_3	$\forall x_7\, ((x_2 + 2) + x_7) = ((x_2 + x_7) + 2)$	$\forall E$ 11
13	a_1, a_2, a_3	$((x_2 + 2) + x_4) = ((x_2 + x_4) + 2)$	$\forall E$ 12
14	a_1, a_2	$x_1 = (x_2 + 2)$	Assume
15	a_1, a_2, a_3	$x_1 = (x_2 + 2)$	Weak 14
16	a_1, a_2, a_3	$2 = ((x_1 + x_4) + 1)$	Assume
17	a_1, a_2, a_3	$2 = (((x_2 + 2) + x_4) + 1)$	$= E$ 15,16
18	a_1, a_2, a_3	$2 = (((x_2 + x_4) + 2) + 1)$	$= E$ 13,17
19	a_1, a_2, a_3	$2 = ((2 + (x_2 + x_4)) + 1)$	$= E$ 8,18
20	a_1, a_2, a_3	$\exists x_3\, 2 = ((2 + x_3) + 1)$	$\exists I$ 19
21	a_1, a_2, a_3	$2 < 2$	$\to E$ 7,20
22	a_1, a_2	$2 < 2$	$\exists E$ 5,21
23	a_1, a_2	$\neg 2 < 2$	Axiom O1
24	a_1	$\neg x_1 = (x_2 + 2)$	$\neg I$ 22,23
25	$\varnothing$	$(x_1 < 2 \to \neg x_1 = (x_2 + 2))$	$\to I$ 24
26	$\varnothing$	$\forall x_2\, (x_1 < 2 \to \neg x_1 = (x_2 + 2))$	$\forall I$ 25
27	$\varnothing$	$\forall x_1\, \forall x_2\, (x_1 < 2 \to \neg x_1 = (x_2 + 2))$	$\forall I$ 26

8.14 double successor

Statement. $\forall x_1\, F_1$.

Variable key. $\texttt{dsD} = x_2$, $\texttt{dsS} = x_3$, $\texttt{dsX} = x_1$, $\texttt{rotX} = x_4$, $\texttt{rotY} = x_5$, $\texttt{rotZ} = x_6$.

Textual formula abbreviations. Expand these from top to bottom; each right-hand side refers only to earlier abbreviations.

F_1 $\quad \forall x_2\, \forall x_3\, ((2 \cdot ((x_1 + x_2) + 1)) + x_3) = (((2 \cdot x_1) + ((2 \cdot x_2) + x_3)) + 2)$

F_2 $\quad \forall x_6\, ((((2 \cdot x_1) + (2 \cdot x_2)) + x_5) + x_6) = ((((2 \cdot x_1) + (2 \cdot x_2)) + x_6) + x_5)$

F_3 $\quad \forall x_6\, ((((2 \cdot x_1) + (2 \cdot x_2)) + 2) + x_6) = ((((2 \cdot x_1) + (2 \cdot x_2)) + x_6) + 2)$

Line	Open assumptions	Formula	Rule and premises
1	$\varnothing$	$(((2 \cdot x_1) + (2 \cdot x_2)) + x_3) = ((2 \cdot x_1) + ((2 \cdot x_2) + x_3))$	Axiom A1
2	$\varnothing$	$\forall x_4\, \forall x_5\, \forall x_6\, ((x_4 + x_5) + x_6) = ((x_4 + x_6) + x_5)$	Lemma 5
3	$\varnothing$	$\forall x_4\, \forall x_5\, \forall x_6\, ((x_4 + x_5) + x_6) = ((x_4 + x_6) + x_5)$	Closed-Weak 2
4	$\varnothing$	$\forall x_5\, F_2$	$\forall E$ 3
5	$\varnothing$	F_3	$\forall E$ 4
6	$\varnothing$	$((((2 \cdot x_1) + (2 \cdot x_2)) + 2) + x_3) = ((((2 \cdot x_1) + (2 \cdot x_2)) + x_3) + 2)$	$\forall E$ 5
7	$\varnothing$	$(2 \cdot 1) = 2$	Axiom A6
8	$\varnothing$	$(2 \cdot (x_1 + x_2)) = ((2 \cdot x_1) + (2 \cdot x_2))$	Axiom A8
9	$\varnothing$	$(2 \cdot ((x_1 + x_2) + 1)) = ((2 \cdot (x_1 + x_2)) + (2 \cdot 1))$	Axiom A8
10	$\varnothing$	$((2 \cdot ((x_1 + x_2) + 1)) + x_3) = ((2 \cdot ((x_1 + x_2) + 1)) + x_3)$	$= I$
11	$\varnothing$	$((2 \cdot ((x_1 + x_2) + 1)) + x_3) = (((2 \cdot (x_1 + x_2)) + (2 \cdot 1)) + x_3)$	$= E$ 9,10
12	$\varnothing$	$((2 \cdot ((x_1 + x_2) + 1)) + x_3) = ((((2 \cdot x_1) + (2 \cdot x_2)) + (2 \cdot 1)) + x_3)$	$= E$ 8,11
13	$\varnothing$	$((2 \cdot ((x_1 + x_2) + 1)) + x_3) = ((((2 \cdot x_1) + (2 \cdot x_2)) + 2) + x_3)$	$= E$ 7,12
14	$\varnothing$	$((2 \cdot ((x_1 + x_2) + 1)) + x_3) = ((((2 \cdot x_1) + (2 \cdot x_2)) + x_3) + 2)$	$= E$ 6,13
15	$\varnothing$	$((2 \cdot ((x_1 + x_2) + 1)) + x_3) = (((2 \cdot x_1) + ((2 \cdot x_2) + x_3)) + 2)$	$= E$ 1,14
16	$\varnothing$	$\forall x_3\, ((2 \cdot ((x_1 + x_2) + 1)) + x_3) = (((2 \cdot x_1) + ((2 \cdot x_2) + x_3)) + 2)$	$\forall I$ 15
17	$\varnothing$	F_1	$\forall I$ 16
18	$\varnothing$	$\forall x_1\, F_1$	$\forall I$ 17

8.15 quotients separated

Statement. $\forall x_1\,\forall x_2\,F_1$.

Variable key. $\texttt{dsD} = x_{11}$, $\texttt{dsS} = x_{12}$, $\texttt{dsX} = x_{10}$, $\texttt{e1} = x_4$, $\texttt{e2} = x_8$, $\texttt{lcX} = x_5$, $\texttt{lcY} = x_6$, $\texttt{lcZ} = x_7$, $\texttt{ntX} = x_{13}$, $\texttt{ntY} = x_{14}$, $\texttt{sepR} = x_3$, $\texttt{sepS} = x_9$, $\texttt{sepX} = x_1$, $\texttt{sepY} = x_2$.

Textual formula abbreviations. Expand these from top to bottom; each right-hand side refers only to earlier abbreviations.

F_1	$\forall x_3\,\forall x_9\,((x_1 < x_2 \wedge x_3 < 2) \to \neg((2 \cdot x_1) + x_3) = ((2 \cdot x_2) + x_9))$
F_2	$(((2 \cdot x_1) + x_3) = ((2 \cdot x_1) + (((2 \cdot x_8) + x_9) + 2)) \to x_3 = (((2 \cdot x_8) + x_9) + 2))$
F_3	$\forall x_{12}\,((2 \cdot ((x_{10} + x_{11}) + 1)) + x_{12}) = (((2 \cdot x_{10}) + ((2 \cdot x_{11}) + x_{12})) + 2)$
F_4	$\forall x_{11}\,\forall x_{12}\,((2 \cdot ((x_1 + x_{11}) + 1)) + x_{12}) = (((2 \cdot x_1) + ((2 \cdot x_{11}) + x_{12})) + 2)$

Assumption key.

a_1	$(x_1 < x_2 \wedge x_3 < 2)$
a_2	$x_2 = ((x_1 + x_8) + 1)$
a_3	$((2 \cdot x_1) + x_3) = ((2 \cdot x_2) + x_9)$

Line	Open assumptions	Formula	Rule and premises
1	a_1	$(x_1 < x_2 \leftrightarrow \exists x_4\,x_2 = ((x_1 + x_4) + 1))$	Axiom O7
2	a_1	$(x_1 < x_2 \to \exists x_4\,x_2 = ((x_1 + x_4) + 1))$	$\leftrightarrow E_1$ 1
3	a_1	$(x_1 < x_2 \wedge x_3 < 2)$	Assume
4	a_1	$x_1 < x_2$	$\wedge E_1$ 3
5	a_1	$\exists x_4\,x_2 = ((x_1 + x_4) + 1)$	$\to E$ 2,4
6	$\varnothing$	$\forall x_5\,\forall x_6\,\forall x_7\,((x_5 + x_6) = (x_5 + x_7) \to x_6 = x_7)$	Lemma 12
7	a_1, a_2, a_3	$\forall x_5\,\forall x_6\,\forall x_7\,((x_5 + x_6) = (x_5 + x_7) \to x_6 = x_7)$	Closed-Weak 6
8	a_1, a_2, a_3	$\forall x_6\,\forall x_7\,(((2 \cdot x_1) + x_6) = ((2 \cdot x_1) + x_7) \to x_6 = x_7)$	$\forall E$ 7
9	a_1, a_2, a_3	$\forall x_7\,(((2 \cdot x_1) + x_3) = ((2 \cdot x_1) + x_7) \to x_3 = x_7)$	$\forall E$ 8
10	a_1, a_2, a_3	F_2	$\forall E$ 9
11	a_1, a_2, a_3	$(((2 \cdot x_1) + ((2 \cdot x_8) + x_9)) + 2) = ((2 \cdot x_1) + (((2 \cdot x_8) + x_9) + 2))$	Axiom A1
12	$\varnothing$	$\forall x_{10}\,\forall x_{11}\,F_3$	Lemma 14
13	a_1, a_2, a_3	$\forall x_{10}\,\forall x_{11}\,F_3$	Closed-Weak 12
14	a_1, a_2, a_3	F_4	$\forall E$ 13
15	a_1, a_2, a_3	$\forall x_{12}\,((2 \cdot ((x_1 + x_8) + 1)) + x_{12}) = (((2 \cdot x_1) + ((2 \cdot x_8) + x_{12})) + 2)$	$\forall E$ 14
16	a_1, a_2, a_3	$((2 \cdot ((x_1 + x_8) + 1)) + x_9) = (((2 \cdot x_1) + ((2 \cdot x_8) + x_9)) + 2)$	$\forall E$ 15
17	a_1, a_2	$x_2 = ((x_1 + x_8) + 1)$	Assume
18	a_1, a_2, a_3	$x_2 = ((x_1 + x_8) + 1)$	Weak 17
19	a_1, a_2, a_3	$((2 \cdot x_1) + x_3) = ((2 \cdot x_2) + x_9)$	Assume
20	a_1, a_2, a_3	$((2 \cdot x_1) + x_3) = ((2 \cdot ((x_1 + x_8) + 1)) + x_9)$	$= E$ 18,19
21	a_1, a_2, a_3	$((2 \cdot x_1) + x_3) = (((2 \cdot x_1) + ((2 \cdot x_8) + x_9)) + 2)$	$= E$ 16,20
22	a_1, a_2, a_3	$((2 \cdot x_1) + x_3) = ((2 \cdot x_1) + (((2 \cdot x_8) + x_9) + 2))$	$= E$ 11,21
23	a_1, a_2, a_3	$x_3 = (((2 \cdot x_8) + x_9) + 2)$	$\to E$ 10,22
24	$\varnothing$	$\forall x_{13}\,\forall x_{14}\,(x_{13} < 2 \to \neg x_{13} = (x_{14} + 2))$	Lemma 13
25	a_1, a_2, a_3	$\forall x_{13}\,\forall x_{14}\,(x_{13} < 2 \to \neg x_{13} = (x_{14} + 2))$	Closed-Weak 24
26	a_1, a_2, a_3	$\forall x_{14}\,(x_3 < 2 \to \neg x_3 = (x_{14} + 2))$	$\forall E$ 25
27	a_1, a_2, a_3	$(x_3 < 2 \to \neg x_3 = (((2 \cdot x_8) + x_9) + 2))$	$\forall E$ 26
28	a_1	$x_3 < 2$	$\wedge E_2$ 3
29	a_1, a_2	$x_3 < 2$	Weak 28
30	a_1, a_2, a_3	$x_3 < 2$	Weak 29
31	a_1, a_2, a_3	$\neg x_3 = (((2 \cdot x_8) + x_9) + 2)$	$\to E$ 27,30
32	a_1, a_2	$\neg((2 \cdot x_1) + x_3) = ((2 \cdot x_2) + x_9)$	$\neg I$ 23,31
33	a_1	$\neg((2 \cdot x_1) + x_3) = ((2 \cdot x_2) + x_9)$	$\exists E$ 5,32
34	$\varnothing$	$((x_1 < x_2 \wedge x_3 < 2) \to \neg((2 \cdot x_1) + x_3) = ((2 \cdot x_2) + x_9))$	$\to I$ 33
35	$\varnothing$	$\forall x_9\,((x_1 < x_2 \wedge x_3 < 2) \to \neg((2 \cdot x_1) + x_3) = ((2 \cdot x_2) + x_9))$	$\forall I$ 34

Line	Open assumptions	Formula	Rule and premises
36	$\varnothing$	F_1	$\forall I$ 35
37	$\varnothing$	$\forall x_2\, F_1$	$\forall I$ 36
38	$\varnothing$	$\forall x_1\, \forall x_2\, F_1$	$\forall I$ 37

8.16 binary pop unique

Statement. $\forall x_1\,\forall x_3\,\forall x_2\,\forall x_4\,F_1$.

Only uniqueness of an already supplied pop is used.

Variable key. $\texttt{buR} = x_2$, $\texttt{buS} = x_4$, $\texttt{buX} = x_1$, $\texttt{buY} = x_3$, $\texttt{lcX} = x_9$, $\texttt{lcY} = x_{10}$, $\texttt{lcZ} = x_{11}$, $\texttt{sepR} = x_7$, $\texttt{sepS} = x_8$, $\texttt{sepX} = x_5$, $\texttt{sepY} = x_6$.

Textual formula abbreviations. Expand these from top to bottom; each right-hand side refers only to earlier abbreviations.

F_1 $((((2 \cdot x_1) + x_2) = ((2 \cdot x_3) + x_4) \wedge (x_2 < 2 \wedge x_4 < 2)) \to (x_1 = x_3 \wedge x_2 = x_4))$

F_2 $\forall x_7\,\forall x_8\,((x_5 < x_6 \wedge x_7 < 2) \to \neg((2 \cdot x_5) + x_7) = ((2 \cdot x_6) + x_8))$

F_3 $\forall x_7\,\forall x_8\,((x_1 < x_6 \wedge x_7 < 2) \to \neg((2 \cdot x_1) + x_7) = ((2 \cdot x_6) + x_8))$

F_4 $\forall x_7\,\forall x_8\,((x_1 < x_3 \wedge x_7 < 2) \to \neg((2 \cdot x_1) + x_7) = ((2 \cdot x_3) + x_8))$

F_5 $\forall x_7\,\forall x_8\,((x_3 < x_6 \wedge x_7 < 2) \to \neg((2 \cdot x_3) + x_7) = ((2 \cdot x_6) + x_8))$

F_6 $\forall x_7\,\forall x_8\,((x_3 < x_1 \wedge x_7 < 2) \to \neg((2 \cdot x_3) + x_7) = ((2 \cdot x_1) + x_8))$

Assumption key.

a_1 $(((2 \cdot x_1) + x_2) = ((2 \cdot x_3) + x_4) \wedge (x_2 < 2 \wedge x_4 < 2))$

a_2 $x_1 < x_3$

a_3 $(x_1 = x_3 \vee x_3 < x_1)$

a_4 $x_1 = x_3$

a_5 $x_3 < x_1$

Line	Open assumptions	Formula	Rule and premises
1	a_1	$(x_1 < x_3 \vee (x_1 = x_3 \vee x_3 < x_1))$	Axiom O3
2	a_1	$(((2 \cdot x_1) + x_2) = ((2 \cdot x_3) + x_4) \wedge (x_2 < 2 \wedge x_4 < 2))$	Assume
3	a_1	$((2 \cdot x_1) + x_2) = ((2 \cdot x_3) + x_4)$	$\wedge E_1$ 2
4	a_1, a_2	$((2 \cdot x_1) + x_2) = ((2 \cdot x_3) + x_4)$	Weak 3
5	$\varnothing$	$\forall x_5\,\forall x_6\,F_2$	Lemma 15
6	a_1, a_2	$\forall x_5\,\forall x_6\,F_2$	Closed-Weak 5
7	a_1, a_2	$\forall x_6\,F_3$	$\forall E$ 6
8	a_1, a_2	F_4	$\forall E$ 7
9	a_1, a_2	$\forall x_8\,((x_1 < x_3 \wedge x_2 < 2) \to \neg((2 \cdot x_1) + x_2) = ((2 \cdot x_3) + x_8))$	$\forall E$ 8
10	a_1, a_2	$((x_1 < x_3 \wedge x_2 < 2) \to \neg((2 \cdot x_1) + x_2) = ((2 \cdot x_3) + x_4))$	$\forall E$ 9
11	a_1, a_2	$x_1 < x_3$	Assume
12	a_1	$(x_2 < 2 \wedge x_4 < 2)$	$\wedge E_2$ 2
13	a_1	$x_2 < 2$	$\wedge E_1$ 12
14	a_1, a_2	$x_2 < 2$	Weak 13
15	a_1, a_2	$(x_1 < x_3 \wedge x_2 < 2)$	$\wedge I$ 11,14
16	a_1, a_2	$\neg((2 \cdot x_1) + x_2) = ((2 \cdot x_3) + x_4)$	$\to E$ 10,15
17	a_1, a_2	$x_1 = x_3$	$\bot E$ 4,16
18	a_1, a_3	$(x_1 = x_3 \vee x_3 < x_1)$	Assume
19	a_1, a_3, a_4	$x_1 = x_3$	Assume
20	a_1, a_3	$((2 \cdot x_1) + x_2) = ((2 \cdot x_3) + x_4)$	Weak 3
21	a_1, a_3, a_5	$((2 \cdot x_1) + x_2) = ((2 \cdot x_3) + x_4)$	Weak 20
22	a_1, a_3, a_5	$((2 \cdot x_3) + x_4) = ((2 \cdot x_1) + x_2)$	= sym 21
23	a_1, a_3, a_5	$\forall x_5\,\forall x_6\,F_2$	Closed-Weak 5
24	a_1, a_3, a_5	$\forall x_6\,F_5$	$\forall E$ 23
25	a_1, a_3, a_5	F_6	$\forall E$ 24
26	a_1, a_3, a_5	$\forall x_8\,((x_3 < x_1 \wedge x_4 < 2) \to \neg((2 \cdot x_3) + x_4) = ((2 \cdot x_1) + x_8))$	$\forall E$ 25
27	a_1, a_3, a_5	$((x_3 < x_1 \wedge x_4 < 2) \to \neg((2 \cdot x_3) + x_4) = ((2 \cdot x_1) + x_2))$	$\forall E$ 26
28	a_1, a_3, a_5	$x_3 < x_1$	Assume
29	a_1	$x_4 < 2$	$\wedge E_2$ 12
30	a_1, a_3	$x_4 < 2$	Weak 29

Line	Open assumptions	Formula	Rule and premises
31	a_1, a_3, a_5	$x_4 < 2$	Weak 30
32	a_1, a_3, a_5	$(x_3 < x_1 \wedge x_4 < 2)$	$\wedge I$ 28,31
33	a_1, a_3, a_5	$\neg((2 \cdot x_3) + x_4) = ((2 \cdot x_1) + x_2)$	$\rightarrow E$ 27,32
34	a_1, a_3, a_5	$x_1 = x_3$	$\bot E$ 22,33
35	a_1, a_3	$x_1 = x_3$	$\vee E$ 18,19,34
36	a_1	$x_1 = x_3$	$\vee E$ 1,17,35
37	$\varnothing$	$\forall x_9 \forall x_{10} \forall x_{11} ((x_9 + x_{10}) = (x_9 + x_{11}) \rightarrow x_{10} = x_{11})$	Lemma 12
38	a_1	$\forall x_9 \forall x_{10} \forall x_{11} ((x_9 + x_{10}) = (x_9 + x_{11}) \rightarrow x_{10} = x_{11})$	Closed-Weak 37
39	a_1	$\forall x_{10} \forall x_{11} (((2 \cdot x_3) + x_{10}) = ((2 \cdot x_3) + x_{11}) \rightarrow x_{10} = x_{11})$	$\forall E$ 38
40	a_1	$\forall x_{11} (((2 \cdot x_3) + x_2) = ((2 \cdot x_3) + x_{11}) \rightarrow x_2 = x_{11})$	$\forall E$ 39
41	a_1	$(((2 \cdot x_3) + x_2) = ((2 \cdot x_3) + x_4) \rightarrow x_2 = x_4)$	$\forall E$ 40
42	a_1	$((2 \cdot x_3) + x_2) = ((2 \cdot x_3) + x_4)$	$= E$ 36,3
43	a_1	$x_2 = x_4$	$\rightarrow E$ 41,42
44	a_1	$(x_1 = x_3 \wedge x_2 = x_4)$	$\wedge I$ 36,43
45	$\varnothing$	F_1	$\rightarrow I$ 44
46	$\varnothing$	$\forall x_4 F_1$	$\forall I$ 45
47	$\varnothing$	$\forall x_2 \forall x_4 F_1$	$\forall I$ 46
48	$\varnothing$	$\forall x_3 \forall x_2 \forall x_4 F_1$	$\forall I$ 47
49	$\varnothing$	$\forall x_1 \forall x_3 \forall x_2 \forall x_4 F_1$	$\forall I$ 48

8.17 exact pattern

Statement. $\forall x_2\, \forall x_1\, (\mathsf{Pat}_{0,x_2}(x_1) \leftrightarrow x_1 = x_2)$.

Variable key. `e2` $= x_4$, `epP` $= x_2$, `epX` $= x_1$, `iu` $= u$, `posX` $= x_5$, `zLeftX` $= x_3$.

Assumption key.

a_1	$\mathsf{Pat}_{0,x_2}(x_1)$
a_2	$(x_4 < (x_1 + 1) \wedge x_1 = ((0 \cdot x_4) + x_2))$
a_3	$x_1 = x_2$

Line	Open assumptions	Formula	Rule and premises
1	a_1	$\mathsf{Pat}_{0,x_2}(x_1)$	Assume
2	$\varnothing$	$\forall x_3\, (0 + x_3) = x_3$	Lemma 1
3	a_1, a_2	$\forall x_3\, (0 + x_3) = x_3$	Closed-Weak 2
4	a_1, a_2	$(0 + x_2) = x_2$	$\forall E$ 3
5	a_1, a_2	$(x_4 \cdot 0) = 0$	Axiom A7
6	a_1, a_2	$(0 \cdot x_4) = (x_4 \cdot 0)$	Axiom A5
7	a_1, a_2	$(x_4 < (x_1 + 1) \wedge x_1 = ((0 \cdot x_4) + x_2))$	Assume
8	a_1, a_2	$x_1 = ((0 \cdot x_4) + x_2)$	$\wedge E_2$ 7
9	a_1, a_2	$x_1 = ((x_4 \cdot 0) + x_2)$	$= E$ 6,8
10	a_1, a_2	$x_1 = (0 + x_2)$	$= E$ 5,9
11	a_1, a_2	$x_1 = x_2$	$= E$ 4,10
12	a_1	$x_1 = x_2$	$\exists E$ 1,11
13	$\varnothing$	$(\mathsf{Pat}_{0,x_2}(x_1) \rightarrow x_1 = x_2)$	$\rightarrow I$ 12
14	$\varnothing$	$\forall x_5\, 0 < (x_5 + 1)$	Lemma 3
15	a_3	$\forall x_5\, 0 < (x_5 + 1)$	Closed-Weak 14
16	a_3	$0 < (x_1 + 1)$	$\forall E$ 15
17	a_3	$(0 \cdot 0) = 0$	Axiom A7
18	a_3	$0 = (0 \cdot 0)$	$=$ sym 17
19	a_3	$\forall x_3\, (0 + x_3) = x_3$	Closed-Weak 2
20	a_3	$(0 + x_2) = x_2$	$\forall E$ 19
21	a_3	$x_2 = (0 + x_2)$	$=$ sym 20
22	a_3	$x_1 = x_2$	Assume
23	a_3	$x_1 = (0 + x_2)$	$= E$ 21,22
24	a_3	$x_1 = ((0 \cdot 0) + x_2)$	$= E$ 18,23
25	a_3	$(0 < (x_1 + 1) \wedge x_1 = ((0 \cdot 0) + x_2))$	$\wedge I$ 16,24
26	a_3	$\mathsf{Pat}_{0,x_2}(x_1)$	$\exists I$ 25
27	$\varnothing$	$(x_1 = x_2 \rightarrow \mathsf{Pat}_{0,x_2}(x_1))$	$\rightarrow I$ 26
28	$\varnothing$	$(\mathsf{Pat}_{0,x_2}(x_1) \leftrightarrow x_1 = x_2)$	$\leftrightarrow I$ 13,27
29	$\varnothing$	$\forall x_1\, (\mathsf{Pat}_{0,x_2}(x_1) \leftrightarrow x_1 = x_2)$	$\forall I$ 28
30	$\varnothing$	$\forall x_2\, \forall x_1\, (\mathsf{Pat}_{0,x_2}(x_1) \leftrightarrow x_1 = x_2)$	$\forall I$ 29

8.18 free pattern

Statement. $\forall x_2\, \mathsf{Pat}_{1,0}(x_2)$.

Variable key. $\texttt{fpX} = x_2$, $\texttt{iu} = u$, $\texttt{succX} = x_1$.

Line	Open assumptions	Formula	Rule and premises
1	$\varnothing$	$\forall x_1\, x_1 < (x_1 + 1)$	Lemma 2
2	$\varnothing$	$\forall x_1\, x_1 < (x_1 + 1)$	Closed-Weak 1
3	$\varnothing$	$x_2 < (x_2 + 1)$	$\forall E$ 2
4	$\varnothing$	$(x_2 \cdot 1) = (1 \cdot x_2)$	Axiom A5
5	$\varnothing$	$(x_2 \cdot 1) = x_2$	Axiom A6
6	$\varnothing$	$x_2 = (x_2 \cdot 1)$	= sym 5
7	$\varnothing$	$(x_2 + 0) = x_2$	Axiom A3
8	$\varnothing$	$x_2 = (x_2 + 0)$	= sym 7
9	$\varnothing$	$x_2 = x_2$	$= I$
10	$\varnothing$	$x_2 = (x_2 + 0)$	$= E$ 8,9
11	$\varnothing$	$x_2 = ((x_2 \cdot 1) + 0)$	$= E$ 6,10
12	$\varnothing$	$x_2 = ((1 \cdot x_2) + 0)$	$= E$ 4,11
13	$\varnothing$	$(x_2 < (x_2 + 1) \wedge x_2 = ((1 \cdot x_2) + 0))$	$\wedge I$ 3,12
14	$\varnothing$	$\mathsf{Pat}_{1,0}(x_2)$	$\exists I$ 13
15	$\varnothing$	$\forall x_2\, \mathsf{Pat}_{1,0}(x_2)$	$\forall I$ 14

8.19 pattern push

Statement. $\forall x_2\,\forall x_3\,\forall x_1\,\forall x_4\,\forall x_5\,F_1$.

Variable key. `e2` $= x_{10}$, `facA` $= x_{11}$, `facB` $= x_{12}$, `facC` $= x_{13}$, `facD` $= x_{14}$, `facE` $= x_{15}$, `iu` $= u$, `ppM` $= x_2$, `ppP` $= x_3$, `ppW` $= x_5$, `ppX` $= x_1$, `ppY` $= x_4$, `trU` $= x_6$, `trW` $= x_9$, `trX` $= x_7$, `trY` $= x_8$.

Textual formula abbreviations. Expand these from top to bottom; each right-hand side refers only to earlier abbreviations.

F_1 $((\mathsf{Pat}_{x_2,x_3}(x_1) \wedge x_4 = ((2 \cdot x_1) + x_5)) \to \mathsf{Pat}_{(2\cdot x_2),((2\cdot x_3)+x_5)}(x_4))$
F_2 $\forall x_8\,\forall x_9\,((x_6 < (x_7 + 1) \wedge x_8 = ((2 \cdot x_7) + x_9)) \to x_6 < (x_8 + 1))$
F_3 $\forall x_8\,\forall x_9\,((x_{10} < (x_7 + 1) \wedge x_8 = ((2 \cdot x_7) + x_9)) \to x_{10} < (x_8 + 1))$
F_4 $\forall x_8\,\forall x_9\,((x_{10} < (x_1 + 1) \wedge x_8 = ((2 \cdot x_1) + x_9)) \to x_{10} < (x_8 + 1))$
F_5 $(((x_{11} \cdot x_{12}) \cdot x_{13}) + ((x_{11} \cdot x_{14}) + x_{15})) = ((x_{11} \cdot ((x_{12} \cdot x_{13}) + x_{14})) + x_{15})$
F_6 $(((2 \cdot x_{12}) \cdot x_{13}) + ((2 \cdot x_{14}) + x_{15})) = ((2 \cdot ((x_{12} \cdot x_{13}) + x_{14})) + x_{15})$
F_7 $(((2 \cdot x_2) \cdot x_{13}) + ((2 \cdot x_{14}) + x_{15})) = ((2 \cdot ((x_2 \cdot x_{13}) + x_{14})) + x_{15})$
F_8 $(((2 \cdot x_2) \cdot x_{10}) + ((2 \cdot x_{14}) + x_{15})) = ((2 \cdot ((x_2 \cdot x_{10}) + x_{14})) + x_{15})$
F_9 $\forall x_{15}\,(((2 \cdot x_2) \cdot x_{10}) + ((2 \cdot x_3) + x_{15})) = ((2 \cdot ((x_2 \cdot x_{10}) + x_3)) + x_{15})$

Assumption key.

a_1 $(\mathsf{Pat}_{x_2,x_3}(x_1) \wedge x_4 = ((2 \cdot x_1) + x_5))$
a_2 $(x_{10} < (x_1 + 1) \wedge x_1 = ((x_2 \cdot x_{10}) + x_3))$

Line	Open assumptions	Formula	Rule and premises
1	a_1	$(\mathsf{Pat}_{x_2,x_3}(x_1) \wedge x_4 = ((2 \cdot x_1) + x_5))$	Assume
2	a_1	$\mathsf{Pat}_{x_2,x_3}(x_1)$	$\wedge E_1$ 1
3	$\varnothing$	$\forall x_6\,\forall x_7\,F_2$	Lemma 11
4	a_1, a_2	$\forall x_6\,\forall x_7\,F_2$	Closed-Weak 3
5	a_1, a_2	$\forall x_7\,F_3$	$\forall E$ 4
6	a_1, a_2	F_4	$\forall E$ 5
7	a_1, a_2	$\forall x_9\,((x_{10} < (x_1{+}1) \wedge x_4 = ((2{\cdot}x_1){+}x_9)) \to x_{10} < (x_4{+}1))$	$\forall E$ 6
8	a_1, a_2	$((x_{10} < (x_1 + 1) \wedge x_4 = ((2 \cdot x_1) + x_5)) \to x_{10} < (x_4 + 1))$	$\forall E$ 7
9	a_1, a_2	$(x_{10} < (x_1 + 1) \wedge x_1 = ((x_2 \cdot x_{10}) + x_3))$	Assume
10	a_1, a_2	$x_{10} < (x_1 + 1)$	$\wedge E_1$ 9
11	a_1	$x_4 = ((2 \cdot x_1) + x_5)$	$\wedge E_2$ 1
12	a_1, a_2	$x_4 = ((2 \cdot x_1) + x_5)$	Weak 11
13	a_1, a_2	$(x_{10} < (x_1 + 1) \wedge x_4 = ((2 \cdot x_1) + x_5))$	$\wedge I$ 10,12
14	a_1, a_2	$x_{10} < (x_4 + 1)$	$\to E$ 8,13
15	$\varnothing$	$\forall x_{11}\,\forall x_{12}\,\forall x_{13}\,\forall x_{14}\,\forall x_{15}\,F_5$	Lemma 7
16	a_1, a_2	$\forall x_{11}\,\forall x_{12}\,\forall x_{13}\,\forall x_{14}\,\forall x_{15}\,F_5$	Closed-Weak 15
17	a_1, a_2	$\forall x_{12}\,\forall x_{13}\,\forall x_{14}\,\forall x_{15}\,F_6$	$\forall E$ 16
18	a_1, a_2	$\forall x_{13}\,\forall x_{14}\,\forall x_{15}\,F_7$	$\forall E$ 17
19	a_1, a_2	$\forall x_{14}\,\forall x_{15}\,F_8$	$\forall E$ 18
20	a_1, a_2	F_9	$\forall E$ 19
21	a_1, a_2	$(((2{\cdot}x_2){\cdot}x_{10}){+}((2{\cdot}x_3){+}x_5)) = ((2{\cdot}((x_2{\cdot}x_{10}){+}x_3)){+}x_5)$	$\forall E$ 20
22	a_1, a_2	$((2{\cdot}((x_2{\cdot}x_{10}){+}x_3)){+}x_5) = (((2{\cdot}x_2){\cdot}x_{10}){+}((2{\cdot}x_3){+}x_5))$	$=$ sym 21
23	a_1, a_2	$x_1 = ((x_2 \cdot x_{10}) + x_3)$	$\wedge E_2$ 9
24	a_1, a_2	$x_4 = ((2 \cdot ((x_2 \cdot x_{10}) + x_3)) + x_5)$	$= E$ 23,12
25	a_1, a_2	$x_4 = (((2 \cdot x_2) \cdot x_{10}) + ((2 \cdot x_3) + x_5))$	$= E$ 22,24
26	a_1, a_2	$(x_{10} < (x_4 + 1) \wedge x_4 = (((2 \cdot x_2) \cdot x_{10}) + ((2 \cdot x_3) + x_5)))$	$\wedge I$ 14,25
27	a_1, a_2	$\mathsf{Pat}_{(2\cdot x_2),((2\cdot x_3)+x_5)}(x_4)$	$\exists I$ 26
28	a_1	$\mathsf{Pat}_{(2\cdot x_2),((2\cdot x_3)+x_5)}(x_4)$	$\exists E$ 2,27
29	$\varnothing$	F_1	$\to I$ 28
30	$\varnothing$	$\forall x_5\,F_1$	$\forall I$ 29
31	$\varnothing$	$\forall x_4\,\forall x_5\,F_1$	$\forall I$ 30
32	$\varnothing$	$\forall x_1\,\forall x_4\,\forall x_5\,F_1$	$\forall I$ 31

Line	Open assumptions	Formula	Rule and premises
33	$\varnothing$	$\forall x_3 \forall x_1 \forall x_4 \forall x_5 F_1$	$\forall I$ 32
34	$\varnothing$	$\forall x_2 \forall x_3 \forall x_1 \forall x_4 \forall x_5 F_1$	$\forall I$ 33

8.20 pattern pop even

Statement. $\forall x_2\,\forall x_3\,\forall x_4\,F_2$.

Variable key. $\mathtt{bdP} = x_{10}$, $\mathtt{bdS} = x_8$, $\mathtt{bdU} = x_9$, $\mathtt{bdX} = x_7$, $\mathtt{buR} = x_{14}$, $\mathtt{buS} = x_{15}$, $\mathtt{buX} = x_{12}$, $\mathtt{buY} = x_{13}$, $\mathtt{e2} = x_{11}$, $\mathtt{facA} = x_{16}$, $\mathtt{facB} = x_{17}$, $\mathtt{facC} = x_{18}$, $\mathtt{facD} = x_{19}$, $\mathtt{facE} = x_{20}$, $\mathtt{iu} = u$, $\mathtt{peE} = x_4$, $\mathtt{peR} = x_6$, $\mathtt{peS} = x_2$, $\mathtt{peV} = x_3$, $\mathtt{peX} = x_1$, $\mathtt{peY} = x_5$.

Textual formula abbreviations. Expand these from top to bottom; each right-hand side refers only to earlier abbreviations.

F_1 $\quad \mathsf{Pat}_{(2\cdot(x_2+1)),((2\cdot x_3)+x_4)}(x_1) \wedge (x_1 = ((2\cdot x_5)+x_6) \wedge (x_6 < 2 \wedge x_4 < 2)))$

F_2 $\quad \forall x_1\,\forall x_5\,\forall x_6\,(F_1 \to (\mathsf{Pat}_{(x_2+1),x_3}(x_5) \wedge x_6 = x_4))$

F_3 $\quad \forall x_8\,\forall x_9\,\forall x_{10}\,(x_7 = (((x_8+1)\cdot x_9)+x_{10}) \to x_9 < (x_7+1))$

F_4 $\quad \forall x_8\,\forall x_9\,\forall x_{10}\,(x_5 = (((x_8+1)\cdot x_9)+x_{10}) \to x_9 < (x_5+1))$

F_5 $\quad ((((2\cdot x_{12})+x_{14}) = ((2\cdot x_{13})+x_{15}) \wedge (x_{14} < 2 \wedge x_{15} < 2)) \to (x_{12} = x_{13} \wedge x_{14} = x_{15}))$

F_6 $\quad ((((2\cdot x_5)+x_{14}) = ((2\cdot x_{13})+x_{15}) \wedge (x_{14} < 2 \wedge x_{15} < 2)) \to (x_5 = x_{13} \wedge x_{14} = x_{15}))$

F_7 $\quad (((2\cdot x_5)+x_{14}) = ((2\cdot (((x_2+1)\cdot x_{11})+x_3))+x_{15}) \wedge (x_{14} < 2 \wedge x_{15} < 2))$

F_8 $\quad \forall x_{14}\,\forall x_{15}\,(F_7 \to (x_5 = (((x_2+1)\cdot x_{11})+x_3) \wedge x_{14} = x_{15}))$

F_9 $\quad (((2\cdot x_5)+x_6) = ((2\cdot (((x_2+1)\cdot x_{11})+x_3))+x_{15}) \wedge (x_6 < 2 \wedge x_{15} < 2))$

F_{10} $\quad (((2\cdot x_5)+x_6) = ((2\cdot (((x_2+1)\cdot x_{11})+x_3))+x_4) \wedge (x_6 < 2 \wedge x_4 < 2))$

F_{11} $\quad (((x_{16}\cdot x_{17})\cdot x_{18})+((x_{16}\cdot x_{19})+x_{20})) = ((x_{16}\cdot ((x_{17}\cdot x_{18})+x_{19}))+x_{20})$

F_{12} $\quad (((2\cdot x_{17})\cdot x_{18})+((2\cdot x_{19})+x_{20})) = ((2\cdot ((x_{17}\cdot x_{18})+x_{19}))+x_{20})$

F_{13} $\quad (((2\cdot (x_2+1))\cdot x_{18})+((2\cdot x_{19})+x_{20})) = ((2\cdot (((x_2+1)\cdot x_{18})+x_{19}))+x_{20})$

F_{14} $\quad (((2\cdot (x_2+1))\cdot x_{11})+((2\cdot x_{19})+x_{20})) = ((2\cdot (((x_2+1)\cdot x_{11})+x_{19}))+x_{20})$

F_{15} $\quad (((2\cdot (x_2+1))\cdot x_{11})+((2\cdot x_3)+x_{20})) = ((2\cdot (((x_2+1)\cdot x_{11})+x_3))+x_{20})$

F_{16} $\quad (((2\cdot (x_2+1))\cdot x_{11})+((2\cdot x_3)+x_4)) = ((2\cdot (((x_2+1)\cdot x_{11})+x_3))+x_4)$

Assumption key.

a_1 $\quad F_1$

a_2 $\quad (x_{11} < (x_1+1) \wedge x_1 = (((2\cdot (x_2+1))\cdot x_{11})+((2\cdot x_3)+x_4)))$

Line	Open assumptions	Formula	Rule and premises
1	a_1	F_1	Assume
2	a_1	$\mathsf{Pat}_{(2\cdot(x_2+1)),((2\cdot x_3)+x_4)}(x_1)$	$\wedge E_1$ 1
3	$\varnothing$	$\forall x_7\,F_3$	Lemma 9
4	a_1, a_2	$\forall x_7\,F_3$	Closed-Weak 3
5	a_1, a_2	F_4	$\forall E$ 4
6	a_1, a_2	$\forall x_9\,\forall x_{10}\,(x_5 = (((x_2+1)\cdot x_9)+x_{10}) \to x_9 < (x_5+1))$	$\forall E$ 5
7	a_1, a_2	$\forall x_{10}\,(x_5 = (((x_2+1)\cdot x_{11})+x_{10}) \to x_{11} < (x_5+1))$	$\forall E$ 6
8	a_1, a_2	$(x_5 = (((x_2+1)\cdot x_{11})+x_3) \to x_{11} < (x_5+1))$	$\forall E$ 7
9	$\varnothing$	$\forall x_{12}\,\forall x_{13}\,\forall x_{14}\,\forall x_{15}\,F_5$	Lemma 16
10	a_1, a_2	$\forall x_{12}\,\forall x_{13}\,\forall x_{14}\,\forall x_{15}\,F_5$	Closed-Weak 9
11	a_1, a_2	$\forall x_{13}\,\forall x_{14}\,\forall x_{15}\,F_6$	$\forall E$ 10
12	a_1, a_2	F_8	$\forall E$ 11
13	a_1, a_2	$\forall x_{15}\,(F_9 \to (x_5 = (((x_2+1)\cdot x_{11})+x_3) \wedge x_6 = x_{15}))$	$\forall E$ 12
14	a_1, a_2	$(F_{10} \to (x_5 = (((x_2+1)\cdot x_{11})+x_3) \wedge x_6 = x_4))$	$\forall E$ 13
15	$\varnothing$	$\forall x_{16}\,\forall x_{17}\,\forall x_{18}\,\forall x_{19}\,\forall x_{20}\,F_{11}$	Lemma 7
16	a_1, a_2	$\forall x_{16}\,\forall x_{17}\,\forall x_{18}\,\forall x_{19}\,\forall x_{20}\,F_{11}$	Closed-Weak 15
17	a_1, a_2	$\forall x_{17}\,\forall x_{18}\,\forall x_{19}\,\forall x_{20}\,F_{12}$	$\forall E$ 16
18	a_1, a_2	$\forall x_{18}\,\forall x_{19}\,\forall x_{20}\,F_{13}$	$\forall E$ 17
19	a_1, a_2	$\forall x_{19}\,\forall x_{20}\,F_{14}$	$\forall E$ 18
20	a_1, a_2	$\forall x_{20}\,F_{15}$	$\forall E$ 19
21	a_1, a_2	F_{16}	$\forall E$ 20
22	a_1	$(x_1 = ((2\cdot x_5)+x_6) \wedge (x_6 < 2 \wedge x_4 < 2))$	$\wedge E_2$ 1
23	a_1	$x_1 = ((2\cdot x_5)+x_6)$	$\wedge E_1$ 22
24	a_1, a_2	$x_1 = ((2\cdot x_5)+x_6)$	Weak 23

Line	Open assumptions	Formula	Rule and premises
25	a_1, a_2	$((2 \cdot x_5) + x_6) = x_1$	= sym 24
26	a_1, a_2	$(x_{11} < (x_1+1) \wedge x_1 = (((2\cdot(x_2+1))\cdot x_{11})+((2\cdot x_3)+x_4)))$	Assume
27	a_1, a_2	$x_1 = (((2 \cdot (x_2 + 1)) \cdot x_{11}) + ((2 \cdot x_3) + x_4))$	$\wedge E_2$ 26
28	a_1, a_2	$((2 \cdot x_5) + x_6) = (((2 \cdot (x_2 + 1)) \cdot x_{11}) + ((2 \cdot x_3) + x_4))$	= trans 25,27
29	a_1, a_2	$((2 \cdot x_5) + x_6) = ((2 \cdot (((x_2 + 1) \cdot x_{11}) + x_3)) + x_4)$	$= E$ 21,28
30	a_1	$(x_6 < 2 \wedge x_4 < 2)$	$\wedge E_2$ 22
31	a_1, a_2	$(x_6 < 2 \wedge x_4 < 2)$	Weak 30
32	a_1, a_2	F_{10}	$\wedge I$ 29,31
33	a_1, a_2	$(x_5 = (((x_2 + 1) \cdot x_{11}) + x_3) \wedge x_6 = x_4)$	$\rightarrow E$ 14,32
34	a_1, a_2	$x_5 = (((x_2 + 1) \cdot x_{11}) + x_3)$	$\wedge E_1$ 33
35	a_1, a_2	$x_{11} < (x_5 + 1)$	$\rightarrow E$ 8,34
36	a_1, a_2	$(x_{11} < (x_5 + 1) \wedge x_5 = (((x_2 + 1) \cdot x_{11}) + x_3))$	$\wedge I$ 35,34
37	a_1, a_2	$\mathsf{Pat}_{(x_2+1),x_3}(x_5)$	$\exists I$ 36
38	a_1, a_2	$x_6 = x_4$	$\wedge E_2$ 33
39	a_1, a_2	$(\mathsf{Pat}_{(x_2+1),x_3}(x_5) \wedge x_6 = x_4)$	$\wedge I$ 37,38
40	a_1	$(\mathsf{Pat}_{(x_2+1),x_3}(x_5) \wedge x_6 = x_4)$	$\exists E$ 2,39
41	$\varnothing$	$(F_1 \rightarrow (\mathsf{Pat}_{(x_2+1),x_3}(x_5) \wedge x_6 = x_4))$	$\rightarrow I$ 40
42	$\varnothing$	$\forall x_6\,(F_1 \rightarrow (\mathsf{Pat}_{(x_2+1),x_3}(x_5) \wedge x_6 = x_4))$	$\forall I$ 41
43	$\varnothing$	$\forall x_5\,\forall x_6\,(F_1 \rightarrow (\mathsf{Pat}_{(x_2+1),x_3}(x_5) \wedge x_6 = x_4))$	$\forall I$ 42
44	$\varnothing$	F_2	$\forall I$ 43
45	$\varnothing$	$\forall x_4\,F_2$	$\forall I$ 44
46	$\varnothing$	$\forall x_3\,\forall x_4\,F_2$	$\forall I$ 45
47	$\varnothing$	$\forall x_2\,\forall x_3\,\forall x_4\,F_2$	$\forall I$ 46

8.21 pattern pop exact

Statement. $\forall x_4\,\forall x_5\,F_2$.

Variable key. $\mathtt{buR} = x_{10}$, $\mathtt{buS} = x_{11}$, $\mathtt{buX} = x_8$, $\mathtt{buY} = x_9$, $\mathtt{epP} = x_1$, $\mathtt{epX} = x_2$, $\mathtt{iu} = u$, $\mathtt{pxE} = x_5$, $\mathtt{pxR} = x_7$, $\mathtt{pxV} = x_4$, $\mathtt{pxX} = x_3$, $\mathtt{pxY} = x_6$.

Textual formula abbreviations. Expand these from top to bottom; each right-hand side refers only to earlier abbreviations.

F_1 $(\mathsf{Pat}_{0,((2\cdot x_4)+x_5)}(x_3) \wedge (x_3 = ((2\cdot x_6)+x_7) \wedge (x_7 < 2 \wedge x_5 < 2)))$

F_2 $\forall x_3\,\forall x_6\,\forall x_7\,(F_1 \to (\mathsf{Pat}_{0,x_4}(x_6) \wedge x_7 = x_5))$

F_3 $((((2\cdot x_8)+x_{10}) = ((2\cdot x_9)+x_{11}) \wedge (x_{10} < 2 \wedge x_{11} < 2)) \to (x_8 = x_9 \wedge x_{10} = x_{11}))$

F_4 $((((2\cdot x_6)+x_{10}) = ((2\cdot x_9)+x_{11}) \wedge (x_{10} < 2 \wedge x_{11} < 2)) \to (x_6 = x_9 \wedge x_{10} = x_{11}))$

F_5 $((((2\cdot x_6)+x_{10}) = ((2\cdot x_4)+x_{11}) \wedge (x_{10} < 2 \wedge x_{11} < 2)) \to (x_6 = x_4 \wedge x_{10} = x_{11}))$

F_6 $((((2\cdot x_6)+x_7) = ((2\cdot x_4)+x_{11}) \wedge (x_7 < 2 \wedge x_{11} < 2)) \to (x_6 = x_4 \wedge x_7 = x_{11}))$

F_7 $((((2\cdot x_6)+x_7) = ((2\cdot x_4)+x_5) \wedge (x_7 < 2 \wedge x_5 < 2)) \to (x_6 = x_4 \wedge x_7 = x_5))$

F_8 $\forall x_2\,(\mathsf{Pat}_{0,((2\cdot x_4)+x_5)}(x_2) \leftrightarrow x_2 = ((2\cdot x_4)+x_5))$

Assumption key.

a_1 F_1

Line	Open assumptions	Formula	Rule and premises
1	$\varnothing$	$\forall x_1\,\forall x_2\,(\mathsf{Pat}_{0,x_1}(x_2) \leftrightarrow x_2 = x_1)$	Lemma 17
2	a_1	$\forall x_1\,\forall x_2\,(\mathsf{Pat}_{0,x_1}(x_2) \leftrightarrow x_2 = x_1)$	Closed-Weak 1
3	a_1	$\forall x_2\,(\mathsf{Pat}_{0,x_4}(x_2) \leftrightarrow x_2 = x_4)$	$\forall E$ 2
4	a_1	$(\mathsf{Pat}_{0,x_4}(x_6) \leftrightarrow x_6 = x_4)$	$\forall E$ 3
5	a_1	$(x_6 = x_4 \to \mathsf{Pat}_{0,x_4}(x_6))$	$\leftrightarrow E_2$ 4
6	$\varnothing$	$\forall x_8\,\forall x_9\,\forall x_{10}\,\forall x_{11}\,F_3$	Lemma 16
7	a_1	$\forall x_8\,\forall x_9\,\forall x_{10}\,\forall x_{11}\,F_3$	Closed-Weak 6
8	a_1	$\forall x_9\,\forall x_{10}\,\forall x_{11}\,F_4$	$\forall E$ 7
9	a_1	$\forall x_{10}\,\forall x_{11}\,F_5$	$\forall E$ 8
10	a_1	$\forall x_{11}\,F_6$	$\forall E$ 9
11	a_1	F_7	$\forall E$ 10
12	a_1	F_1	Assume
13	a_1	$(x_3 = ((2\cdot x_6)+x_7) \wedge (x_7 < 2 \wedge x_5 < 2))$	$\wedge E_2$ 12
14	a_1	$x_3 = ((2\cdot x_6)+x_7)$	$\wedge E_1$ 13
15	a_1	$((2\cdot x_6)+x_7) = x_3$	= sym 14
16	a_1	F_8	$\forall E$ 2
17	a_1	$(\mathsf{Pat}_{0,((2\cdot x_4)+x_5)}(x_3) \leftrightarrow x_3 = ((2\cdot x_4)+x_5))$	$\forall E$ 16
18	a_1	$(\mathsf{Pat}_{0,((2\cdot x_4)+x_5)}(x_3) \to x_3 = ((2\cdot x_4)+x_5))$	$\leftrightarrow E_1$ 17
19	a_1	$\mathsf{Pat}_{0,((2\cdot x_4)+x_5)}(x_3)$	$\wedge E_1$ 12
20	a_1	$x_3 = ((2\cdot x_4)+x_5)$	$\to E$ 18,19
21	a_1	$((2\cdot x_6)+x_7) = ((2\cdot x_4)+x_5)$	= trans 15,20
22	a_1	$(x_7 < 2 \wedge x_5 < 2)$	$\wedge E_2$ 13
23	a_1	$(((2\cdot x_6)+x_7) = ((2\cdot x_4)+x_5) \wedge (x_7 < 2 \wedge x_5 < 2))$	$\wedge I$ 21,22
24	a_1	$(x_6 = x_4 \wedge x_7 = x_5)$	$\to E$ 11,23
25	a_1	$x_6 = x_4$	$\wedge E_1$ 24
26	a_1	$\mathsf{Pat}_{0,x_4}(x_6)$	$\to E$ 5,25
27	a_1	$x_7 = x_5$	$\wedge E_2$ 24
28	a_1	$(\mathsf{Pat}_{0,x_4}(x_6) \wedge x_7 = x_5)$	$\wedge I$ 26,27
29	$\varnothing$	$(F_1 \to (\mathsf{Pat}_{0,x_4}(x_6) \wedge x_7 = x_5))$	$\to I$ 28
30	$\varnothing$	$\forall x_7\,(F_1 \to (\mathsf{Pat}_{0,x_4}(x_6) \wedge x_7 = x_5))$	$\forall I$ 29
31	$\varnothing$	$\forall x_6\,\forall x_7\,(F_1 \to (\mathsf{Pat}_{0,x_4}(x_6) \wedge x_7 = x_5))$	$\forall I$ 30
32	$\varnothing$	F_2	$\forall I$ 31
33	$\varnothing$	$\forall x_5\,F_2$	$\forall I$ 32

Line	Open assumptions	Formula	Rule and premises
34	$\varnothing$	$\forall x_4 \forall x_5 F_2$	$\forall I$ 33

8.22 pattern inclusion

Statement. $\forall x_2\,\forall x_3\,\forall x_4\,\forall x_5\,\forall x_1\,F_1$.

Variable key. $\mathtt{bdP} = x_9$, $\mathtt{bdS} = x_7$, $\mathtt{bdU} = x_8$, $\mathtt{bdX} = x_6$, $\mathtt{e2} = x_{10}$, $\mathtt{facA} = x_{11}$, $\mathtt{facB} = x_{12}$, $\mathtt{facC} = x_{13}$, $\mathtt{facD} = x_{14}$, $\mathtt{facE} = x_{15}$, $\mathtt{iu} = u$, $\mathtt{piK} = x_3$, $\mathtt{piR} = x_5$, $\mathtt{piS} = x_2$, $\mathtt{piV} = x_4$, $\mathtt{piX} = x_1$.

Textual formula abbreviations. Expand these from top to bottom; each right-hand side refers only to earlier abbreviations.

F_1	$(\mathsf{Pat}_{((x_2+1)\cdot x_3),(((x_2+1)\cdot x_4)+x_5)}(x_1) \to \mathsf{Pat}_{(x_2+1),x_5}(x_1))$
F_2	$\forall x_7\,\forall x_8\,\forall x_9\,(x_6 = (((x_7+1)\cdot x_8)+x_9) \to x_8 < (x_6+1))$
F_3	$\forall x_7\,\forall x_8\,\forall x_9\,(x_1 = (((x_7+1)\cdot x_8)+x_9) \to x_8 < (x_1+1))$
F_4	$(x_1 = (((x_2+1)\cdot((x_3\cdot x_{10})+x_4))+x_9) \to ((x_3\cdot x_{10})+x_4) < (x_1+1))$
F_5	$(x_1 = (((x_2+1)\cdot((x_3\cdot x_{10})+x_4))+x_5) \to ((x_3\cdot x_{10})+x_4) < (x_1+1))$
F_6	$(((x_{11}\cdot x_{12})\cdot x_{13})+((x_{11}\cdot x_{14})+x_{15})) = ((x_{11}\cdot((x_{12}\cdot x_{13})+x_{14}))+x_{15})$
F_7	$((((x_2+1)\cdot x_{12})\cdot x_{13})+(((x_2+1)\cdot x_{14})+x_{15})) = (((x_2+1)\cdot((x_{12}\cdot x_{13})+x_{14}))+x_{15})$
F_8	$((((x_2+1)\cdot x_3)\cdot x_{13})+(((x_2+1)\cdot x_{14})+x_{15})) = (((x_2+1)\cdot((x_3\cdot x_{13})+x_{14}))+x_{15})$
F_9	$((((x_2+1)\cdot x_3)\cdot x_{10})+(((x_2+1)\cdot x_{14})+x_{15})) = (((x_2+1)\cdot((x_3\cdot x_{10})+x_{14}))+x_{15})$
F_{10}	$((((x_2+1)\cdot x_3)\cdot x_{10})+(((x_2+1)\cdot x_4)+x_{15})) = (((x_2+1)\cdot((x_3\cdot x_{10})+x_4))+x_{15})$
F_{11}	$((((x_2+1)\cdot x_3)\cdot x_{10})+(((x_2+1)\cdot x_4)+x_5)) = (((x_2+1)\cdot((x_3\cdot x_{10})+x_4))+x_5)$
F_{12}	$(((x_3\cdot x_{10})+x_4) < (x_1+1) \wedge x_1 = (((x_2+1)\cdot((x_3\cdot x_{10})+x_4))+x_5))$

Assumption key.

a_1	$\mathsf{Pat}_{((x_2+1)\cdot x_3),(((x_2+1)\cdot x_4)+x_5)}(x_1)$
a_2	$(x_{10} < (x_1+1) \wedge x_1 = ((((x_2+1)\cdot x_3)\cdot x_{10})+(((x_2+1)\cdot x_4)+x_5)))$

Line	Open assumptions	Formula	Rule and premises
1	a_1	$\mathsf{Pat}_{((x_2+1)\cdot x_3),(((x_2+1)\cdot x_4)+x_5)}(x_1)$	Assume
2	$\varnothing$	$\forall x_6\,F_2$	Lemma 9
3	a_1, a_2	$\forall x_6\,F_2$	Closed-Weak 2
4	a_1, a_2	F_3	$\forall E$ 3
5	a_1, a_2	$\forall x_8\,\forall x_9\,(x_1 = (((x_2+1)\cdot x_8)+x_9) \to x_8 < (x_1+1))$	$\forall E$ 4
6	a_1, a_2	$\forall x_9\,F_4$	$\forall E$ 5
7	a_1, a_2	F_5	$\forall E$ 6
8	$\varnothing$	$\forall x_{11}\,\forall x_{12}\,\forall x_{13}\,\forall x_{14}\,\forall x_{15}\,F_6$	Lemma 7
9	a_1, a_2	$\forall x_{11}\,\forall x_{12}\,\forall x_{13}\,\forall x_{14}\,\forall x_{15}\,F_6$	Closed-Weak 8
10	a_1, a_2	$\forall x_{12}\,\forall x_{13}\,\forall x_{14}\,\forall x_{15}\,F_7$	$\forall E$ 9
11	a_1, a_2	$\forall x_{13}\,\forall x_{14}\,\forall x_{15}\,F_8$	$\forall E$ 10
12	a_1, a_2	$\forall x_{14}\,\forall x_{15}\,F_9$	$\forall E$ 11
13	a_1, a_2	$\forall x_{15}\,F_{10}$	$\forall E$ 12
14	a_1, a_2	F_{11}	$\forall E$ 13
15	a_1, a_2	$(x_{10} < (x_1+1) \wedge x_1 = (((((x_2+1)\cdot x_3)\cdot x_{10})+(((x_2+1)\cdot x_4)+x_5)))$	Assume
16	a_1, a_2	$x_1 = ((((x_2+1)\cdot x_3)\cdot x_{10})+(((x_2+1)\cdot x_4)+x_5))$	$\wedge E_2$ 15
17	a_1, a_2	$x_1 = (((x_2+1)\cdot((x_3\cdot x_{10})+x_4))+x_5)$	$=E$ 14,16
18	a_1, a_2	$((x_3\cdot x_{10})+x_4) < (x_1+1)$	$\to E$ 7,17
19	a_1, a_2	F_{12}	$\wedge I$ 18,17
20	a_1, a_2	$\mathsf{Pat}_{(x_2+1),x_5}(x_1)$	$\exists I$ 19
21	a_1	$\mathsf{Pat}_{(x_2+1),x_5}(x_1)$	$\exists E$ 1,20
22	$\varnothing$	F_1	$\to I$ 21
23	$\varnothing$	$\forall x_1\,F_1$	$\forall I$ 22
24	$\varnothing$	$\forall x_5\,\forall x_1\,F_1$	$\forall I$ 23
25	$\varnothing$	$\forall x_4\,\forall x_5\,\forall x_1\,F_1$	$\forall I$ 24
26	$\varnothing$	$\forall x_3\,\forall x_4\,\forall x_5\,\forall x_1\,F_1$	$\forall I$ 25
27	$\varnothing$	$\forall x_2\,\forall x_3\,\forall x_4\,\forall x_5\,\forall x_1\,F_1$	$\forall I$ 26

8.23 bit bound

Statement. $\forall x_1\,((x_1 = 0 \lor x_1 = 1) \to x_1 < 2)$.

Variable key. $\texttt{bitX} = x_1$, $\texttt{posX} = x_2$, $\texttt{succX} = x_3$.

Assumption key.

a_1	$(x_1 = 0 \lor x_1 = 1)$
a_2	$x_1 = 0$
a_3	$x_1 = 1$

Line	Open assumptions	Formula	Rule and premises
1	a_1	$(x_1 = 0 \lor x_1 = 1)$	Assume
2	a_1, a_2	$x_1 = 0$	Assume
3	a_1, a_2	$0 = x_1$	$=$ sym 2
4	$\varnothing$	$\forall x_2\, 0 < (x_2 + 1)$	Lemma 3
5	a_1, a_2	$\forall x_2\, 0 < (x_2 + 1)$	Closed-Weak 4
6	a_1, a_2	$0 < 2$	$\forall E$ 5
7	a_1, a_2	$x_1 < 2$	$= E$ 3,6
8	a_1, a_3	$x_1 = 1$	Assume
9	a_1, a_3	$1 = x_1$	$=$ sym 8
10	$\varnothing$	$\forall x_3\, x_3 < (x_3 + 1)$	Lemma 2
11	a_1, a_3	$\forall x_3\, x_3 < (x_3 + 1)$	Closed-Weak 10
12	a_1, a_3	$1 < 2$	$\forall E$ 11
13	a_1, a_3	$x_1 < 2$	$= E$ 9,12
14	a_1	$x_1 < 2$	$\lor E$ 1,7,13
15	$\varnothing$	$((x_1 = 0 \lor x_1 = 1) \to x_1 < 2)$	$\to I$ 14
16	$\varnothing$	$\forall x_1\,((x_1 = 0 \lor x_1 = 1) \to x_1 < 2)$	$\forall I$ 15

8.24 curry

Statement. $(((\varphi \wedge \psi) \to \chi) \to (\varphi \to (\psi \to \chi)))$.

Assumption key.

a_1	$((\varphi \wedge \psi) \to \chi)$
a_2	φ
a_3	ψ

Line	Open assumptions	Formula	Rule and premises
1	a_1	$((\varphi \wedge \psi) \to \chi)$	Assume
2	a_1, a_2	$((\varphi \wedge \psi) \to \chi)$	Weak 1
3	a_1, a_2, a_3	$((\varphi \wedge \psi) \to \chi)$	Weak 2
4	a_1, a_2	φ	Assume
5	a_1, a_2, a_3	φ	Weak 4
6	a_1, a_2, a_3	ψ	Assume
7	a_1, a_2, a_3	$(\varphi \wedge \psi)$	$\wedge I$ 5,6
8	a_1, a_2, a_3	χ	$\to E$ 3,7
9	a_1, a_2	$(\psi \to \chi)$	$\to I$ 8
10	a_1	$(\varphi \to (\psi \to \chi))$	$\to I$ 9
11	$\varnothing$	$(((\varphi \wedge \psi) \to \chi) \to (\varphi \to (\psi \to \chi)))$	$\to I$ 10

8.25 or fusion

Statement. F_1.

Textual formula abbreviations. Expand these from top to bottom; each right-hand side refers only to earlier abbreviations.

F_1	$(((\varphi \to (\psi \to \chi)) \land (\varphi \to (\theta \to \chi))) \to (\varphi \to ((\psi \lor \theta) \to \chi)))$

Assumption key.

a_1	$((\varphi \to (\psi \to \chi)) \land (\varphi \to (\theta \to \chi)))$
a_2	φ
a_3	$(\psi \lor \theta)$
a_4	ψ
a_5	θ

Context key. The lists retain assumption order.

Γ_0	a_1, a_2, a_3
Γ_1	a_1
Γ_2	a_1, a_2
Γ_3	a_1, a_2, a_3, a_4
Γ_4	a_1, a_2, a_3, a_5
Γ_5	$\varnothing$

Line	Open assumptions	Formula	Rule and premises
1	Γ_0	$(\psi \lor \theta)$	Assume
2	Γ_1	$((\varphi \to (\psi \to \chi)) \land (\varphi \to (\theta \to \chi)))$	Assume
3	Γ_1	$(\varphi \to (\psi \to \chi))$	$\land E_1$ 2
4	Γ_2	$(\varphi \to (\psi \to \chi))$	Weak 3
5	Γ_2	φ	Assume
6	Γ_2	$(\psi \to \chi)$	$\to E$ 4,5
7	Γ_0	$(\psi \to \chi)$	Weak 6
8	Γ_3	$(\psi \to \chi)$	Weak 7
9	Γ_3	ψ	Assume
10	Γ_3	χ	$\to E$ 8,9
11	Γ_1	$(\varphi \to (\theta \to \chi))$	$\land E_2$ 2
12	Γ_2	$(\varphi \to (\theta \to \chi))$	Weak 11
13	Γ_2	$(\theta \to \chi)$	$\to E$ 12,5
14	Γ_0	$(\theta \to \chi)$	Weak 13
15	Γ_4	$(\theta \to \chi)$	Weak 14
16	Γ_4	θ	Assume
17	Γ_4	χ	$\to E$ 15,16
18	Γ_0	χ	$\lor E$ 1,10,17
19	Γ_2	$((\psi \lor \theta) \to \chi)$	$\to I$ 18
20	Γ_1	$(\varphi \to ((\psi \lor \theta) \to \chi))$	$\to I$ 19
21	Γ_5	F_1	$\to I$ 20

8.26 equality clash

Statement. $((t = u \wedge (t = v \wedge \neg u = v)) \rightarrow \chi)$.

Variable key. $\mathtt{t} = t$, $\mathtt{u} = u$, $\mathtt{v} = v$.

Assumption key.

a_1 $(t = u \wedge (t = v \wedge \neg u = v))$

Line	Open assumptions	Formula	Rule and premises
1	a_1	$(t = u \wedge (t = v \wedge \neg u = v))$	Assume
2	a_1	$t = u$	$\wedge E_1$ 1
3	a_1	$u = t$	$=$ sym 2
4	a_1	$(t = v \wedge \neg u = v)$	$\wedge E_2$ 1
5	a_1	$t = v$	$\wedge E_1$ 4
6	a_1	$u = v$	$=$ trans 3,5
7	a_1	$\neg u = v$	$\wedge E_2$ 4
8	a_1	χ	$\perp E$ 6,7
9	$\varnothing$	$((t = u \wedge (t = v \wedge \neg u = v)) \rightarrow \chi)$	$\rightarrow I$ 8

8.27 strict not equal

Statement. $\forall x_1 \, \forall x_2 \, (x_1 < x_2 \to \neg x_1 = x_2)$.

Variable key. `neqX` $= x_1$, `neqY` $= x_2$.

Assumption key.

a_1	$x_1 < x_2$
a_2	$x_1 = x_2$

Line	Open assumptions	Formula	Rule and premises
1	a_1, a_2	$x_1 = x_2$	Assume
2	a_1, a_2	$x_2 = x_1$	$=$ sym 1
3	a_1	$x_1 < x_2$	Assume
4	a_1, a_2	$x_1 < x_2$	Weak 3
5	a_1, a_2	$x_1 < x_1$	$= E$ 2,4
6	a_1, a_2	$\neg x_1 < x_1$	Axiom O1
7	a_1	$\neg x_1 = x_2$	$\neg I$ 5,6
8	$\varnothing$	$(x_1 < x_2 \to \neg x_1 = x_2)$	$\to I$ 7
9	$\varnothing$	$\forall x_2 \, (x_1 < x_2 \to \neg x_1 = x_2)$	$\forall I$ 8
10	$\varnothing$	$\forall x_1 \, \forall x_2 \, (x_1 < x_2 \to \neg x_1 = x_2)$	$\forall I$ 9

8.28 zero pattern

Statement. $\forall x_2\, \mathsf{Pat}_{x_2,0}(0)$.

Variable key. $\texttt{iu} = u$, $\texttt{posX} = x_1$, $\texttt{zeroPatternM} = x_2$.

Line	Open assumptions	Formula	Rule and premises
1	$\varnothing$	$\forall x_1\, 0 < (x_1 + 1)$	Lemma 3
2	$\varnothing$	$\forall x_1\, 0 < (x_1 + 1)$	Closed-Weak 1
3	$\varnothing$	$0 < (0 + 1)$	$\forall E$ 2
4	$\varnothing$	$(x_2 \cdot 0) = 0$	Axiom A7
5	$\varnothing$	$0 = (x_2 \cdot 0)$	$=$ sym 4
6	$\varnothing$	$(0 + 0) = 0$	Axiom A3
7	$\varnothing$	$0 = (0 + 0)$	$=$ sym 6
8	$\varnothing$	$0 = 0$	$= I$
9	$\varnothing$	$0 = (0 + 0)$	$= E$ 7,8
10	$\varnothing$	$0 = ((x_2 \cdot 0) + 0)$	$= E$ 5,9
11	$\varnothing$	$(0 < (0 + 1) \wedge 0 = ((x_2 \cdot 0) + 0))$	$\wedge I$ 3,10
12	$\varnothing$	$\mathsf{Pat}_{x_2,0}(0)$	$\exists I$ 11
13	$\varnothing$	$\forall x_2\, \mathsf{Pat}_{x_2,0}(0)$	$\forall I$ 12

8.29 lt add cancel

Statement. $\forall x_1\,\forall x_3\,\forall x_2\,((x_1 + x_2) < (x_3 + x_2) \to x_1 < x_3)$.

Variable key. $\texttt{ordX} = x_1$, $\texttt{ordY} = x_3$, $\texttt{ordZ} = x_2$.

Assumption key.

a_1	$(x_1 + x_2) < (x_3 + x_2)$
a_2	$x_1 < x_3$
a_3	$(x_1 = x_3 \vee x_3 < x_1)$
a_4	$x_1 = x_3$
a_5	$x_3 < x_1$

Line	Open assumptions	Formula	Rule and premises
1	a_1	$(x_1 < x_3 \vee (x_1 = x_3 \vee x_3 < x_1))$	Axiom O3
2	a_1, a_2	$x_1 < x_3$	Assume
3	a_1, a_3	$(x_1 = x_3 \vee x_3 < x_1)$	Assume
4	a_1, a_3, a_4	$x_1 = x_3$	Assume
5	a_1, a_3, a_4	$x_2 = x_2$	$= I$
6	a_1, a_3, a_4	$(x_1 + x_2) = (x_3 + x_2)$	+cong 4,5
7	a_1	$(x_1 + x_2) < (x_3 + x_2)$	Assume
8	a_1, a_3	$(x_1 + x_2) < (x_3 + x_2)$	Weak 7
9	a_1, a_3, a_4	$(x_1 + x_2) < (x_3 + x_2)$	Weak 8
10	a_1, a_3, a_4	$(x_3 + x_2) < (x_3 + x_2)$	$= E$ 6,9
11	a_1, a_3, a_4	$\neg(x_3 + x_2) < (x_3 + x_2)$	Axiom O1
12	a_1, a_3, a_4	$x_1 < x_3$	$\perp E$ 10,11
13	a_1, a_3, a_5	$(((x_3 + x_2) < (x_1 + x_2) \wedge (x_1 + x_2) < (x_3 + x_2)) \to (x_3 + x_2) < (x_3 + x_2))$	Axiom O2
14	a_1, a_3, a_5	$(x_3 < x_1 \to (x_3 + x_2) < (x_1 + x_2))$	Axiom O5
15	a_1, a_3, a_5	$x_3 < x_1$	Assume
16	a_1, a_3, a_5	$(x_3 + x_2) < (x_1 + x_2)$	$\to E$ 14,15
17	a_1, a_3, a_5	$(x_1 + x_2) < (x_3 + x_2)$	Weak 8
18	a_1, a_3, a_5	$((x_3 + x_2) < (x_1 + x_2) \wedge (x_1 + x_2) < (x_3 + x_2))$	$\wedge I$ 16,17
19	a_1, a_3, a_5	$(x_3 + x_2) < (x_3 + x_2)$	$\to E$ 13,18
20	a_1, a_3, a_5	$\neg(x_3 + x_2) < (x_3 + x_2)$	Axiom O1
21	a_1, a_3, a_5	$x_1 < x_3$	$\perp E$ 19,20
22	a_1, a_3	$x_1 < x_3$	$\vee E$ 3,12,21
23	a_1	$x_1 < x_3$	$\vee E$ 1,2,22
24	$\varnothing$	$((x_1 + x_2) < (x_3 + x_2) \to x_1 < x_3)$	$\to I$ 23
25	$\varnothing$	$\forall x_2\,((x_1 + x_2) < (x_3 + x_2) \to x_1 < x_3)$	$\forall I$ 24
26	$\varnothing$	$\forall x_3\,\forall x_2\,((x_1 + x_2) < (x_3 + x_2) \to x_1 < x_3)$	$\forall I$ 25
27	$\varnothing$	$\forall x_1\,\forall x_3\,\forall x_2\,((x_1 + x_2) < (x_3 + x_2) \to x_1 < x_3)$	$\forall I$ 26

8.30 bounded history invariant

Statement. $((\mathsf{Init} \wedge (\mathsf{Close} \wedge (\mathsf{Final} \wedge \mathsf{Run}))) \to g_0 < g_1)$.

One bounded induction instance; the goal is the displayed atomic inequality.

Variable key. $\mathtt{B0} = B_0$, $\mathtt{B1} = B_1$, $\mathtt{B2} = B_2$, $\mathtt{B3} = B_3$, $\mathtt{C0} = C_0$, $\mathtt{C1} = C_1$, $\mathtt{C2} = C_2$, $\mathtt{C3} = C_3$, $\mathtt{bu} = v$, $\mathtt{cb} = x_{42}$, $\mathtt{cert0b} = a_0$, $\mathtt{cert0flag} = f_0$, $\mathtt{cert0lm} = m_0$, $\mathtt{cert0lp} = p_0$, $\mathtt{cert0q} = s_0$, $\mathtt{cert0rm} = r_0$, $\mathtt{cert0rp} = z_0$, $\mathtt{cert0time} = \tau_0$, $\mathtt{cert1b} = a_1$, $\mathtt{cert1flag} = f_1$, $\mathtt{cert1lm} = m_1$, $\mathtt{cert1lp} = p_1$, $\mathtt{cert1q} = s_1$, $\mathtt{cert1rm} = r_1$, $\mathtt{cert1rp} = z_1$, $\mathtt{cert1time} = \tau_1$, $\mathtt{cert2b} = a_2$, $\mathtt{cert2flag} = f_2$, $\mathtt{cert2lm} = m_2$, $\mathtt{cert2lp} = p_2$, $\mathtt{cert2q} = s_2$, $\mathtt{cert2rm} = r_2$, $\mathtt{cert2rp} = z_2$, $\mathtt{cert2time} = \tau_2$, $\mathtt{cert3b} = a_3$, $\mathtt{cert3flag} = f_3$, $\mathtt{cert3lm} = m_3$, $\mathtt{cert3lp} = p_3$, $\mathtt{cert3q} = s_3$, $\mathtt{cert3rm} = r_3$, $\mathtt{cert3rp} = z_3$, $\mathtt{cert3time} = \tau_3$, $\mathtt{cert4b} = a_4$, $\mathtt{cert4flag} = f_4$, $\mathtt{cert4lm} = m_4$, $\mathtt{cert4lp} = p_4$, $\mathtt{cert4q} = s_4$, $\mathtt{cert4rm} = r_4$, $\mathtt{cert4rp} = z_4$, $\mathtt{cert4time} = \tau_4$, $\mathtt{cert5b} = a_5$, $\mathtt{cert5flag} = f_5$, $\mathtt{cert5lm} = m_5$, $\mathtt{cert5lp} = p_5$, $\mathtt{cert5q} = s_5$, $\mathtt{cert5rm} = r_5$, $\mathtt{cert5rp} = z_5$, $\mathtt{cert5time} = \tau_5$, $\mathtt{cert6b} = a_6$, $\mathtt{cert6flag} = f_6$, $\mathtt{cert6lm} = m_6$, $\mathtt{cert6lp} = p_6$, $\mathtt{cert6q} = s_6$, $\mathtt{cert6rm} = r_6$, $\mathtt{cert6rp} = z_6$, $\mathtt{cert6time} = \tau_6$, $\mathtt{cert7b} = a_7$, $\mathtt{cert7flag} = f_7$, $\mathtt{cert7lm} = m_7$, $\mathtt{cert7lp} = p_7$, $\mathtt{cert7q} = s_7$, $\mathtt{cert7rm} = r_7$, $\mathtt{cert7rp} = z_7$, $\mathtt{cert7time} = \tau_7$, $\mathtt{cl} = x_{41}$, $\mathtt{cq} = x_{40}$, $\mathtt{cr} = x_{43}$, $\mathtt{d0} = d_0$, $\mathtt{d1} = d_1$, $\mathtt{d2} = d_2$, $\mathtt{d3} = d_3$, $\mathtt{fb} = x_{38}$, $\mathtt{finalHi} = g_1$, $\mathtt{finalLo} = g_0$, $\mathtt{fl} = x_{37}$, $\mathtt{fq} = x_{36}$, $\mathtt{fr} = x_{39}$, $\mathtt{gateCur0} = x_6$, $\mathtt{gateCur1} = x_7$, $\mathtt{gateCur2} = x_8$, $\mathtt{gateCur3} = x_9$, $\mathtt{gateFinal0} = x_{14}$, $\mathtt{gateFinal1} = x_{15}$, $\mathtt{gateFinal2} = x_{16}$, $\mathtt{gateFinal3} = x_{17}$, $\mathtt{gateInit0} = x_1$, $\mathtt{gateInit1} = x_2$, $\mathtt{gateInit2} = x_3$, $\mathtt{gateInit3} = x_4$, $\mathtt{gateNext0} = x_{10}$, $\mathtt{gateNext1} = x_{11}$, $\mathtt{gateNext2} = x_{12}$, $\mathtt{gateNext3} = x_{13}$, $\mathtt{gateTime} = x_5$, $\mathtt{ib} = x_{25}$, $\mathtt{il} = x_{24}$, $\mathtt{iq} = x_{23}$, $\mathtt{ir} = x_{26}$, $\mathtt{iu} = u$, $\mathtt{iv} = i$, $\mathtt{mh} = H$, $\mathtt{mn} = n$, $\mathtt{nb} = x_{34}$, $\mathtt{nl} = x_{33}$, $\mathtt{nq} = x_{32}$, $\mathtt{nr} = x_{35}$, $\mathtt{ob} = x_{46}$, $\mathtt{ol} = x_{45}$, $\mathtt{oq} = x_{44}$, $\mathtt{ordX} = x_{52}$, $\mathtt{ordY} = x_{53}$, $\mathtt{ordZ} = x_{54}$, $\mathtt{orr} = x_{47}$, $\mathtt{sb} = x_{30}$, $\mathtt{si} = x_{27}$, $\mathtt{sl} = x_{29}$, $\mathtt{sq} = x_{28}$, $\mathtt{sr} = x_{31}$, $\mathtt{succX} = x_{55}$, $\mathtt{tb} = x_{21}$, $\mathtt{ti} = x_{18}$, $\mathtt{tl} = x_{20}$, $\mathtt{tq} = x_{19}$, $\mathtt{tr} = x_{22}$, $\mathtt{vb} = x_{50}$, $\mathtt{vl} = x_{49}$, $\mathtt{vq} = x_{48}$, $\mathtt{vr} = x_{51}$.

Textual formula abbreviations. Expand these from top to bottom; each right-hand side refers only to earlier abbreviations.

F_{1} $\exists x_{37} < H\, \exists x_{38} < H\, \exists x_{39} < H\, (\mathsf{At}(n, x_{40}, x_{37}, x_{38}, x_{39}) \wedge x_{40} = 2)$

F_{2} $(x_{41} < H \wedge \exists x_{38} < H\, \exists x_{39} < H\, (\mathsf{At}(n, x_{40}, x_{41}, x_{38}, x_{39}) \wedge x_{40} = 2))$

F_{3} $\forall x_{51} < H\, (\mathsf{At}((i+1), x_{48}, x_{49}, x_{50}, x_{51}) \to I((i+1), x_{48}, x_{49}, x_{50}, x_{51}))$

F_{4} $(x_{40} < H \wedge \exists x_{20} < H\, \exists x_{21} < H\, \exists x_{22} < H\, \mathsf{At}(i, x_{40}, x_{20}, x_{21}, x_{22}))$

F_{5} $(\mathsf{At}(0, x_{23}, x_{24}, x_{25}, x_{26}) \to (x_{23} = 0 \wedge (x_{24} = 0 \wedge (x_{25} = 0 \wedge x_{26} = 0))))$

F_{6} $(\mathsf{At}(x_{27}, x_{28}, x_{29}, x_{30}, x_{31}) \wedge \mathsf{At}((x_{27}+1), x_{32}, x_{33}, x_{34}, x_{35}))$

F_{7} $\forall x_{34} < H\, \forall x_{35} < H\, (F_6 \to \mathsf{Step}(x_{28}, x_{29}, x_{30}, x_{31}, x_{32}, x_{33}, x_{34}, x_{35}))$

F_{8} $\forall x_{28} < H\, \forall x_{29} < H\, \forall x_{30} < H\, \forall x_{31} < H\, \forall x_{32} < H\, \forall x_{33} < H\, F_7$

F_{9} $\exists x_{37} < H\, \exists x_{38} < H\, \exists x_{39} < H\, (\mathsf{At}(n, x_{36}, x_{37}, x_{38}, x_{39}) \wedge x_{36} = 2)$

F_{10} $(\forall x_{23} < H\, \forall x_{24} < H\, \forall x_{25} < H\, \forall x_{26} < H\, F_5 \wedge (\forall x_{27} < n\, F_8 \wedge \exists x_{36} < H\, F_9))$

F_{11} $\forall x_{15} < H\, \forall x_{16} < H\, \forall x_{17} < H\, ((I(n, x_{40}, x_{15}, x_{16}, x_{17}) \wedge x_{40} = 2) \to g_0 < g_1)$

F_{12} $(x_{41} < H \to \forall x_{16} < H\, \forall x_{17} < H\, ((I(n, x_{40}, x_{41}, x_{16}, x_{17}) \wedge x_{40} = 2) \to g_0 < g_1))$

F_{13} $\forall x_{46} < H\, \forall x_{47} < H\, (\mathsf{At}(i, x_{44}, x_{45}, x_{46}, x_{47}) \to I(i, x_{44}, x_{45}, x_{46}, x_{47}))$

F_{14} $((\varphi(0) \wedge \forall i\, (\varphi(i) \to \varphi((i+1)))) \to \forall i < (n+1)\, \forall x_{44} < H\, \forall x_{45} < H\, F_{13})$

F_{15} $\forall x_4 < H\, ((x_{44} = 0 \wedge (x_2 = 0 \wedge (x_3 = 0 \wedge x_4 = 0))) \to I(0, x_{44}, x_2, x_3, x_4))$

F_{16} $\forall x_4 < H\, ((x_{44} = 0 \wedge (x_{45} = 0 \wedge (x_3 = 0 \wedge x_4 = 0))) \to I(0, x_{44}, x_{45}, x_3, x_4))$

F_{17} $\forall x_4 < H\, ((x_{44} = 0 \wedge (x_{45} = 0 \wedge (x_{46} = 0 \wedge x_4 = 0))) \to I(0, x_{44}, x_{45}, x_{46}, x_4))$

F_{18} $(x_{47} < H \to ((x_{44} = 0 \wedge (x_{45} = 0 \wedge (x_{46} = 0 \wedge x_{47} = 0))) \to I(0, x_{44}, x_{45}, x_{46}, x_{47})))$

F_{19} $(\mathsf{At}(0, x_{44}, x_{24}, x_{25}, x_{26}) \to (x_{44} = 0 \wedge (x_{24} = 0 \wedge (x_{25} = 0 \wedge x_{26} = 0))))$

F_{20} $(\mathsf{At}(0, x_{44}, x_{45}, x_{25}, x_{26}) \to (x_{44} = 0 \wedge (x_{45} = 0 \wedge (x_{25} = 0 \wedge x_{26} = 0))))$

F_{21} $(\mathsf{At}(0, x_{44}, x_{45}, x_{46}, x_{26}) \to (x_{44} = 0 \wedge (x_{45} = 0 \wedge (x_{46} = 0 \wedge x_{26} = 0))))$

F_{22} $(\mathsf{At}(0, x_{44}, x_{45}, x_{46}, x_{47}) \to (x_{44} = 0 \wedge (x_{45} = 0 \wedge (x_{46} = 0 \wedge x_{47} = 0))))$

F_{23} $(x_{46} < H \to \forall x_{47} < H\, (\mathsf{At}(0, x_{44}, x_{45}, x_{46}, x_{47}) \to I(0, x_{44}, x_{45}, x_{46}, x_{47})))$

F_{24} $\forall x_{46} < H\, \forall x_{47} < H\, (\mathsf{At}(0, x_{44}, x_{45}, x_{46}, x_{47}) \to I(0, x_{44}, x_{45}, x_{46}, x_{47}))$

F_{25} $\forall x_{51} < H\, (\mathsf{At}((i+1), x_{44}, x_{49}, x_{50}, x_{51}) \to I((i+1), x_{44}, x_{49}, x_{50}, x_{51}))$

F_{26} $\forall x_{51} < H\, (\mathsf{At}((i+1), x_{44}, x_{45}, x_{50}, x_{51}) \to I((i+1), x_{44}, x_{45}, x_{50}, x_{51}))$

F_{27} $\forall x_{51} < H\, (\mathsf{At}((i+1), x_{44}, x_{45}, x_{46}, x_{51}) \to I((i+1), x_{44}, x_{45}, x_{46}, x_{51}))$

F_{28} $(x_{47} < H \to (\mathsf{At}((i+1), x_{44}, x_{45}, x_{46}, x_{47}) \to I((i+1), x_{44}, x_{45}, x_{46}, x_{47})))$

F_{29} $\forall x_{47} < H\, (\mathsf{At}((i+1), x_{44}, x_{45}, x_{46}, x_{47}) \to I((i+1), x_{44}, x_{45}, x_{46}, x_{47}))$

F_{30} $(\forall x_{48} < H\, \forall x_{49} < H\, \forall x_{50} < H\, F_3 \to \forall x_{44} < H\, \forall x_{45} < H\, \forall x_{46} < H\, F_{29})$

F_{31} $\exists x_{20} < H\, \exists x_{21} < H\, \exists x_{22} < H\, \mathsf{At}(x_{18}, x_{19}, x_{20}, x_{21}, x_{22})$

F_{32} $\exists x_{19} < H\, \exists x_{20} < H\, \exists x_{21} < H\, \exists x_{22} < H\, \mathsf{At}(i, x_{19}, x_{20}, x_{21}, x_{22})$

F_{33} $\forall x_{52}\,\forall x_{53}\,\forall x_{54}\,((x_{52}+x_{54})<(x_{53}+x_{54})\to x_{52}<x_{53})$
F_{34} $(I(i,x_6,x_7,x_8,x_9)\wedge \mathsf{Step}(x_6,x_7,x_8,x_9,x_{10},x_{11},x_{12},x_{13}))$
F_{35} $\forall x_{11}<H\,\forall x_{12}<H\,\forall x_{13}<H\,(F_{34}\to I((i+1),x_{10},x_{11},x_{12},x_{13}))$
F_{36} $(I(i,x_{40},x_7,x_8,x_9)\wedge \mathsf{Step}(x_{40},x_7,x_8,x_9,x_{10},x_{11},x_{12},x_{13}))$
F_{37} $\forall x_{11}<H\,\forall x_{12}<H\,\forall x_{13}<H\,(F_{36}\to I((i+1),x_{10},x_{11},x_{12},x_{13}))$
F_{38} $(I(i,x_{40},x_{41},x_8,x_9)\wedge \mathsf{Step}(x_{40},x_{41},x_8,x_9,x_{10},x_{11},x_{12},x_{13}))$
F_{39} $\forall x_{11}<H\,\forall x_{12}<H\,\forall x_{13}<H\,(F_{38}\to I((i+1),x_{10},x_{11},x_{12},x_{13}))$
F_{40} $(I(i,x_{40},x_{41},x_{42},x_9)\wedge \mathsf{Step}(x_{40},x_{41},x_{42},x_9,x_{10},x_{11},x_{12},x_{13}))$
F_{41} $\forall x_{11}<H\,\forall x_{12}<H\,\forall x_{13}<H\,(F_{40}\to I((i+1),x_{10},x_{11},x_{12},x_{13}))$
F_{42} $(I(i,x_{40},x_{41},x_{42},x_{43})\wedge \mathsf{Step}(x_{40},x_{41},x_{42},x_{43},x_{10},x_{11},x_{12},x_{13}))$
F_{43} $\forall x_{11}<H\,\forall x_{12}<H\,\forall x_{13}<H\,(F_{42}\to I((i+1),x_{10},x_{11},x_{12},x_{13}))$
F_{44} $(I(i,x_{40},x_{41},x_{42},x_{43})\wedge \mathsf{Step}(x_{40},x_{41},x_{42},x_{43},x_{48},x_{11},x_{12},x_{13}))$
F_{45} $\forall x_{11}<H\,\forall x_{12}<H\,\forall x_{13}<H\,(F_{44}\to I((i+1),x_{48},x_{11},x_{12},x_{13}))$
F_{46} $(I(i,x_{40},x_{41},x_{42},x_{43})\wedge \mathsf{Step}(x_{40},x_{41},x_{42},x_{43},x_{48},x_{49},x_{12},x_{13}))$
F_{47} $(I(i,x_{40},x_{41},x_{42},x_{43})\wedge \mathsf{Step}(x_{40},x_{41},x_{42},x_{43},x_{48},x_{49},x_{50},x_{13}))$
F_{48} $(I(i,x_{40},x_{41},x_{42},x_{43})\wedge \mathsf{Step}(x_{40},x_{41},x_{42},x_{43},x_{48},x_{49},x_{50},x_{51}))$
F_{49} $\forall x_{46}<H\,\forall x_{47}<H\,(\mathsf{At}(i,x_{40},x_{45},x_{46},x_{47})\to I(i,x_{40},x_{45},x_{46},x_{47}))$
F_{50} $\forall x_{46}<H\,\forall x_{47}<H\,(\mathsf{At}(i,x_{40},x_{41},x_{46},x_{47})\to I(i,x_{40},x_{41},x_{46},x_{47}))$
F_{51} $(x_{42}<H\to\forall x_{47}<H\,(\mathsf{At}(i,x_{40},x_{41},x_{42},x_{47})\to I(i,x_{40},x_{41},x_{42},x_{47})))$
F_{52} $((\mathsf{At}(i,x_{28},x_{29},x_{30},x_{31})\wedge \mathsf{At}((i+1),x_{32},x_{33},x_{34},x_{35}))\to \mathsf{Step}(x_{28},x_{29},x_{30},x_{31},x_{32},x_{33},x_{34},x_{35}))$
F_{53} $\forall x_{31}<H\,\forall x_{32}<H\,\forall x_{33}<H\,\forall x_{34}<H\,\forall x_{35}<H\,F_{52}$
F_{54} $((\mathsf{At}(i,x_{40},x_{29},x_{30},x_{31})\wedge \mathsf{At}((i+1),x_{32},x_{33},x_{34},x_{35}))\to \mathsf{Step}(x_{40},x_{29},x_{30},x_{31},x_{32},x_{33},x_{34},x_{35}))$
F_{55} $\forall x_{31}<H\,\forall x_{32}<H\,\forall x_{33}<H\,\forall x_{34}<H\,\forall x_{35}<H\,F_{54}$
F_{56} $((\mathsf{At}(i,x_{40},x_{41},x_{30},x_{31})\wedge \mathsf{At}((i+1),x_{32},x_{33},x_{34},x_{35}))\to \mathsf{Step}(x_{40},x_{41},x_{30},x_{31},x_{32},x_{33},x_{34},x_{35}))$
F_{57} $\forall x_{31}<H\,\forall x_{32}<H\,\forall x_{33}<H\,\forall x_{34}<H\,\forall x_{35}<H\,F_{56}$
F_{58} $((\mathsf{At}(i,x_{40},x_{41},x_{42},x_{31})\wedge \mathsf{At}((i+1),x_{32},x_{33},x_{34},x_{35}))\to \mathsf{Step}(x_{40},x_{41},x_{42},x_{31},x_{32},x_{33},x_{34},x_{35}))$
F_{59} $\forall x_{31}<H\,\forall x_{32}<H\,\forall x_{33}<H\,\forall x_{34}<H\,\forall x_{35}<H\,F_{58}$
F_{60} $((\mathsf{At}(i,x_{40},x_{41},x_{42},x_{43})\wedge \mathsf{At}((i+1),x_{32},x_{33},x_{34},x_{35}))\to \mathsf{Step}(x_{40},x_{41},x_{42},x_{43},x_{32},x_{33},x_{34},x_{35}))$
F_{61} $((\mathsf{At}(i,x_{40},x_{41},x_{42},x_{43})\wedge \mathsf{At}((i+1),x_{48},x_{33},x_{34},x_{35}))\to \mathsf{Step}(x_{40},x_{41},x_{42},x_{43},x_{48},x_{33},x_{34},x_{35}))$
F_{62} $((\mathsf{At}(i,x_{40},x_{41},x_{42},x_{43})\wedge \mathsf{At}((i+1),x_{48},x_{49},x_{34},x_{35}))\to \mathsf{Step}(x_{40},x_{41},x_{42},x_{43},x_{48},x_{49},x_{34},x_{35}))$
F_{63} $((\mathsf{At}(i,x_{40},x_{41},x_{42},x_{43})\wedge \mathsf{At}((i+1),x_{48},x_{49},x_{50},x_{35}))\to \mathsf{Step}(x_{40},x_{41},x_{42},x_{43},x_{48},x_{49},x_{50},x_{35}))$
F_{64} $((\mathsf{At}(i,x_{40},x_{41},x_{42},x_{43})\wedge \mathsf{At}((i+1),x_{48},x_{49},x_{50},x_{51}))\to \mathsf{Step}(x_{40},x_{41},x_{42},x_{43},x_{48},x_{49},x_{50},x_{51}))$
F_{65} $(x_{51}<H\to(\mathsf{At}((i+1),x_{48},x_{49},x_{50},x_{51})\to I((i+1),x_{48},x_{49},x_{50},x_{51})))$
F_{66} $\forall x_{46}<H\,\forall x_{47}<H\,(\mathsf{At}(n,x_{44},x_{45},x_{46},x_{47})\to I(n,x_{44},x_{45},x_{46},x_{47}))$
F_{67} $\forall x_{46}<H\,\forall x_{47}<H\,(\mathsf{At}(n,x_{40},x_{45},x_{46},x_{47})\to I(n,x_{40},x_{45},x_{46},x_{47}))$
F_{68} $\forall x_{46}<H\,\forall x_{47}<H\,(\mathsf{At}(n,x_{40},x_{41},x_{46},x_{47})\to I(n,x_{40},x_{41},x_{46},x_{47}))$
F_{69} $(x_{42}<H\to\forall x_{47}<H\,(\mathsf{At}(n,x_{40},x_{41},x_{42},x_{47})\to I(n,x_{40},x_{41},x_{42},x_{47})))$

Assumption key.

a_1 $(\mathsf{Init}\wedge(\mathsf{Close}\wedge(\mathsf{Final}\wedge\mathsf{Run})))$
a_2 $(x_{40}<H\wedge F_1)$
a_3 F_2
a_4 $(x_{42}<H\wedge\exists x_{39}<H\,(\mathsf{At}(n,x_{40},x_{41},x_{42},x_{39})\wedge x_{40}=2))$
a_5 $(x_{43}<H\wedge(\mathsf{At}(n,x_{40},x_{41},x_{42},x_{43})\wedge x_{40}=2))$
a_6 $0<(n+1)$
a_7 $x_{44}<H$
a_8 $x_{45}<H$
a_9 $x_{46}<H$
a_{10} $x_{47}<H$
a_{11} $\mathsf{At}(0,x_{44},x_{45},x_{46},x_{47})$
a_{12} $\forall x_{48}<H\,\forall x_{49}<H\,\forall x_{50}<H\,F_3$
a_{13} $\varphi(i)$
a_{14} $(i+1)<(n+1)$
a_{15} $x_{48}<H$
a_{16} $x_{49}<H$

a_{17}	$x_{50} < H$
a_{18}	$x_{51} < H$
a_{19}	$\mathsf{At}((i+1), x_{48}, x_{49}, x_{50}, x_{51})$
a_{20}	F_4
a_{21}	$(x_{41} < H \wedge \exists x_{21} < H \, \exists x_{22} < H \, \mathsf{At}(i, x_{40}, x_{41}, x_{21}, x_{22}))$
a_{22}	$(x_{42} < H \wedge \exists x_{22} < H \, \mathsf{At}(i, x_{40}, x_{41}, x_{42}, x_{22}))$
a_{23}	$(x_{43} < H \wedge \mathsf{At}(i, x_{40}, x_{41}, x_{42}, x_{43}))$

Context key. The lists retain assumption order.

Γ_0	a_1
Γ_1	a_1, a_2
Γ_2	a_1, a_2, a_3
Γ_3	a_1, a_2, a_3, a_4
Γ_4	a_1, a_2, a_3, a_4, a_5
Γ_5	a_1, a_6
Γ_6	a_1, a_6, a_7
Γ_7	a_1, a_6, a_7, a_8
Γ_8	a_1, a_6, a_7, a_8, a_9
Γ_9	$a_1, a_6, a_7, a_8, a_9, a_{10}$
Γ_{10}	$a_1, a_6, a_7, a_8, a_9, a_{10}, a_{11}$
Γ_{11}	a_{12}
Γ_{12}	a_{12}, a_7
Γ_{13}	a_{12}, a_7, a_8
Γ_{14}	a_{12}, a_7, a_8, a_9
Γ_{15}	$a_{12}, a_7, a_8, a_9, a_{10}$
Γ_{16}	$\varnothing$
Γ_{17}	a_1, a_{13}, a_{14}
Γ_{18}	a_1, a_{13}
Γ_{19}	$a_1, a_{13}, a_{14}, a_{15}$
Γ_{20}	$a_1, a_{13}, a_{14}, a_{15}, a_{16}$
Γ_{21}	$a_1, a_{13}, a_{14}, a_{15}, a_{16}, a_{17}$
Γ_{22}	$a_1, a_{13}, a_{14}, a_{15}, a_{16}, a_{17}, a_{18}$
Γ_{23}	$a_1, a_{13}, a_{14}, a_{15}, a_{16}, a_{17}, a_{18}, a_{19}$
Γ_{24}	$a_1, a_{13}, a_{14}, a_{15}, a_{16}, a_{17}, a_{18}, a_{19}, a_{20}$
Γ_{25}	$a_1, a_{13}, a_{14}, a_{15}, a_{16}, a_{17}, a_{18}, a_{19}, a_{20}, a_{21}$
Γ_{26}	$a_1, a_{13}, a_{14}, a_{15}, a_{16}, a_{17}, a_{18}, a_{19}, a_{20}, a_{21}, a_{22}$
Γ_{27}	$a_1, a_{13}, a_{14}, a_{15}, a_{16}, a_{17}, a_{18}, a_{19}, a_{20}, a_{21}, a_{22}, a_{23}$

Line	Open assumptions	Formula	Rule and premises
1	Γ_0	$(\mathsf{Init} \wedge (\mathsf{Close} \wedge (\mathsf{Final} \wedge \mathsf{Run})))$	Assume
2	Γ_0	$(\mathsf{Close} \wedge (\mathsf{Final} \wedge \mathsf{Run}))$	$\wedge E_2$ 1
3	Γ_0	$(\mathsf{Final} \wedge \mathsf{Run})$	$\wedge E_2$ 2
4	Γ_0	Run	$\wedge E_2$ 3
5	Γ_0	F_{10}	$\wedge E_2$ 4
6	Γ_0	$(\forall x_{27} < n \, F_8 \wedge \exists x_{36} < H \, F_9)$	$\wedge E_2$ 5
7	Γ_0	$\exists x_{36} < H \, F_9$	$\wedge E_2$ 6
8	Γ_1	$(x_{40} < H \wedge F_1)$	Assume
9	Γ_1	F_1	$\wedge E_2$ 8
10	Γ_2	F_2	Assume
11	Γ_2	$\exists x_{38} < H \, \exists x_{39} < H \, (\mathsf{At}(n, x_{40}, x_{41}, x_{38}, x_{39}) \wedge x_{40} = 2)$	$\wedge E_2$ 10
12	Γ_3	$(x_{42} < H \wedge \exists x_{39} < H \, (\mathsf{At}(n, x_{40}, x_{41}, x_{42}, x_{39}) \wedge x_{40} = 2))$	Assume
13	Γ_3	$\exists x_{39} < H \, (\mathsf{At}(n, x_{40}, x_{41}, x_{42}, x_{39}) \wedge x_{40} = 2)$	$\wedge E_2$ 12
14	Γ_0	Final	$\wedge E_1$ 3
15	Γ_1	Final	Weak 14

Line	Open assumptions	Formula	Rule and premises
16	Γ_2	Final	Weak 15
17	Γ_3	Final	Weak 16
18	Γ_4	Final	Weak 17
19	Γ_4	$(x_{40} < H \to F_{11})$	$\forall E$ 18
20	Γ_1	$x_{40} < H$	$\wedge E_1$ 8
21	Γ_2	$x_{40} < H$	Weak 20
22	Γ_3	$x_{40} < H$	Weak 21
23	Γ_4	$x_{40} < H$	Weak 22
24	Γ_4	F_{11}	$\to E$ 19,23
25	Γ_4	F_{12}	$\forall E$ 24
26	Γ_2	$x_{41} < H$	$\wedge E_1$ 10
27	Γ_3	$x_{41} < H$	Weak 26
28	Γ_4	$x_{41} < H$	Weak 27
29	Γ_4	$\forall x_{16} < H\, \forall x_{17} < H\, ((I(n, x_{40}, x_{41}, x_{16}, x_{17}) \wedge x_{40} = 2) \to g_0 < g_1)$	$\to E$ 25,28
30	Γ_4	$(x_{42} < H \to \forall x_{17} < H\, ((I(n, x_{40}, x_{41}, x_{42}, x_{17}) \wedge x_{40} = 2) \to g_0 < g_1))$	$\forall E$ 29
31	Γ_3	$x_{42} < H$	$\wedge E_1$ 12
32	Γ_4	$x_{42} < H$	Weak 31
33	Γ_4	$\forall x_{17} < H\, ((I(n, x_{40}, x_{41}, x_{42}, x_{17}) \wedge x_{40} = 2) \to g_0 < g_1)$	$\to E$ 30,32
34	Γ_4	$(x_{43} < H \to ((I(n, x_{40}, x_{41}, x_{42}, x_{43}) \wedge x_{40} = 2) \to g_0 < g_1))$	$\forall E$ 33
35	Γ_4	$(x_{43} < H \wedge (\mathsf{At}(n, x_{40}, x_{41}, x_{42}, x_{43}) \wedge x_{40} = 2))$	Assume
36	Γ_4	$x_{43} < H$	$\wedge E_1$ 35
37	Γ_4	$((I(n, x_{40}, x_{41}, x_{42}, x_{43}) \wedge x_{40} = 2) \to g_0 < g_1)$	$\to E$ 34,36
38	Γ_0	F_{14}	$I\Delta_0$ instance
39	Γ_0	Init	$\wedge E_1$ 1
40	Γ_5	Init	Weak 39
41	Γ_6	Init	Weak 40
42	Γ_7	Init	Weak 41
43	Γ_8	Init	Weak 42
44	Γ_9	Init	Weak 43
45	Γ_{10}	Init	Weak 44
46	Γ_{10}	$(x_{44} < H \to \forall x_2 < H\, \forall x_3 < H\, F_{15})$	$\forall E$ 45
47	Γ_6	$x_{44} < H$	Assume
48	Γ_7	$x_{44} < H$	Weak 47
49	Γ_8	$x_{44} < H$	Weak 48
50	Γ_9	$x_{44} < H$	Weak 49
51	Γ_{10}	$x_{44} < H$	Weak 50
52	Γ_{10}	$\forall x_2 < H\, \forall x_3 < H\, F_{15}$	$\to E$ 46,51
53	Γ_{10}	$(x_{45} < H \to \forall x_3 < H\, F_{16})$	$\forall E$ 52
54	Γ_7	$x_{45} < H$	Assume
55	Γ_8	$x_{45} < H$	Weak 54
56	Γ_9	$x_{45} < H$	Weak 55
57	Γ_{10}	$x_{45} < H$	Weak 56
58	Γ_{10}	$\forall x_3 < H\, F_{16}$	$\to E$ 53,57
59	Γ_{10}	$(x_{46} < H \to F_{17})$	$\forall E$ 58
60	Γ_8	$x_{46} < H$	Assume
61	Γ_9	$x_{46} < H$	Weak 60
62	Γ_{10}	$x_{46} < H$	Weak 61
63	Γ_{10}	F_{17}	$\to E$ 59,62
64	Γ_{10}	F_{18}	$\forall E$ 63
65	Γ_9	$x_{47} < H$	Assume

Line	Open assumptions	Formula	Rule and premises
66	Γ_{10}	$x_{47} < H$	Weak 65
67	Γ_{10}	$((x_{44} = 0 \wedge (x_{45} = 0 \wedge (x_{46} = 0 \wedge x_{47} = 0))) \rightarrow I(0, x_{44}, x_{45}, x_{46}, x_{47}))$	$\rightarrow E$ 64,66
68	Γ_0	$\forall x_{23} < H\, \forall x_{24} < H\, \forall x_{25} < H\, \forall x_{26} < H\, F_5$	$\wedge E_1$ 5
69	Γ_5	$\forall x_{23} < H\, \forall x_{24} < H\, \forall x_{25} < H\, \forall x_{26} < H\, F_5$	Weak 68
70	Γ_6	$\forall x_{23} < H\, \forall x_{24} < H\, \forall x_{25} < H\, \forall x_{26} < H\, F_5$	Weak 69
71	Γ_7	$\forall x_{23} < H\, \forall x_{24} < H\, \forall x_{25} < H\, \forall x_{26} < H\, F_5$	Weak 70
72	Γ_8	$\forall x_{23} < H\, \forall x_{24} < H\, \forall x_{25} < H\, \forall x_{26} < H\, F_5$	Weak 71
73	Γ_9	$\forall x_{23} < H\, \forall x_{24} < H\, \forall x_{25} < H\, \forall x_{26} < H\, F_5$	Weak 72
74	Γ_{10}	$\forall x_{23} < H\, \forall x_{24} < H\, \forall x_{25} < H\, \forall x_{26} < H\, F_5$	Weak 73
75	Γ_{10}	$(x_{44} < H \rightarrow \forall x_{24} < H\, \forall x_{25} < H\, \forall x_{26} < H\, F_{19})$	$\forall E$ 74
76	Γ_{10}	$\forall x_{24} < H\, \forall x_{25} < H\, \forall x_{26} < H\, F_{19}$	$\rightarrow E$ 75,51
77	Γ_{10}	$(x_{45} < H \rightarrow \forall x_{25} < H\, \forall x_{26} < H\, F_{20})$	$\forall E$ 76
78	Γ_{10}	$\forall x_{25} < H\, \forall x_{26} < H\, F_{20}$	$\rightarrow E$ 77,57
79	Γ_{10}	$(x_{46} < H \rightarrow \forall x_{26} < H\, F_{21})$	$\forall E$ 78
80	Γ_{10}	$\forall x_{26} < H\, F_{21}$	$\rightarrow E$ 79,62
81	Γ_{10}	$(x_{47} < H \rightarrow F_{22})$	$\forall E$ 80
82	Γ_{10}	F_{22}	$\rightarrow E$ 81,66
83	Γ_{10}	$\mathsf{At}(0, x_{44}, x_{45}, x_{46}, x_{47})$	Assume
84	Γ_{10}	$(x_{44} = 0 \wedge (x_{45} = 0 \wedge (x_{46} = 0 \wedge x_{47} = 0)))$	$\rightarrow E$ 82,83
85	Γ_{10}	$I(0, x_{44}, x_{45}, x_{46}, x_{47})$	$\rightarrow E$ 67,84
86	Γ_9	$(\mathsf{At}(0, x_{44}, x_{45}, x_{46}, x_{47}) \rightarrow I(0, x_{44}, x_{45}, x_{46}, x_{47}))$	$\rightarrow I$ 85
87	Γ_8	$(x_{47} < H \rightarrow (\mathsf{At}(0, x_{44}, x_{45}, x_{46}, x_{47}) \rightarrow I(0, x_{44}, x_{45}, x_{46}, x_{47})))$	$\rightarrow I$ 86
88	Γ_8	$\forall x_{47} < H\, (\mathsf{At}(0, x_{44}, x_{45}, x_{46}, x_{47}) \rightarrow I(0, x_{44}, x_{45}, x_{46}, x_{47}))$	$\forall I$ 87
89	Γ_7	F_{23}	$\rightarrow I$ 88
90	Γ_7	F_{24}	$\forall I$ 89
91	Γ_6	$(x_{45} < H \rightarrow F_{24})$	$\rightarrow I$ 90
92	Γ_6	$\forall x_{45} < H\, F_{24}$	$\forall I$ 91
93	Γ_5	$(x_{44} < H \rightarrow \forall x_{45} < H\, F_{24})$	$\rightarrow I$ 92
94	Γ_5	$\forall x_{44} < H\, \forall x_{45} < H\, F_{24}$	$\forall I$ 93
95	Γ_0	$\varphi(0)$	$\rightarrow I$ 94
96	Γ_{11}	$\forall x_{48} < H\, \forall x_{49} < H\, \forall x_{50} < H\, F_3$	Assume
97	Γ_{12}	$\forall x_{48} < H\, \forall x_{49} < H\, \forall x_{50} < H\, F_3$	Weak 96
98	Γ_{13}	$\forall x_{48} < H\, \forall x_{49} < H\, \forall x_{50} < H\, F_3$	Weak 97
99	Γ_{14}	$\forall x_{48} < H\, \forall x_{49} < H\, \forall x_{50} < H\, F_3$	Weak 98
100	Γ_{15}	$\forall x_{48} < H\, \forall x_{49} < H\, \forall x_{50} < H\, F_3$	Weak 99
101	Γ_{15}	$(x_{44} < H \rightarrow \forall x_{49} < H\, \forall x_{50} < H\, F_{25})$	$\forall E$ 100
102	Γ_{12}	$x_{44} < H$	Assume
103	Γ_{13}	$x_{44} < H$	Weak 102
104	Γ_{14}	$x_{44} < H$	Weak 103
105	Γ_{15}	$x_{44} < H$	Weak 104
106	Γ_{15}	$\forall x_{49} < H\, \forall x_{50} < H\, F_{25}$	$\rightarrow E$ 101,105
107	Γ_{15}	$(x_{45} < H \rightarrow \forall x_{50} < H\, F_{26})$	$\forall E$ 106
108	Γ_{13}	$x_{45} < H$	Assume
109	Γ_{14}	$x_{45} < H$	Weak 108
110	Γ_{15}	$x_{45} < H$	Weak 109
111	Γ_{15}	$\forall x_{50} < H\, F_{26}$	$\rightarrow E$ 107,110
112	Γ_{15}	$(x_{46} < H \rightarrow F_{27})$	$\forall E$ 111
113	Γ_{14}	$x_{46} < H$	Assume
114	Γ_{15}	$x_{46} < H$	Weak 113
115	Γ_{15}	F_{27}	$\rightarrow E$ 112,114

Line	Open assumptions	Formula	Rule and premises
116	Γ_{15}	F_{28}	$\forall E$ 115
117	Γ_{15}	$x_{47} < H$	Assume
118	Γ_{15}	$(\mathsf{At}((i+1), x_{44}, x_{45}, x_{46}, x_{47}) \rightarrow I((i+1), x_{44}, x_{45}, x_{46}, x_{47}))$	$\rightarrow E$ 116,117
119	Γ_{14}	F_{28}	$\rightarrow I$ 118
120	Γ_{14}	F_{29}	$\forall I$ 119
121	Γ_{13}	$(x_{46} < H \rightarrow F_{29})$	$\rightarrow I$ 120
122	Γ_{13}	$\forall x_{46} < H\, F_{29}$	$\forall I$ 121
123	Γ_{12}	$(x_{45} < H \rightarrow \forall x_{46} < H\, F_{29})$	$\rightarrow I$ 122
124	Γ_{12}	$\forall x_{45} < H\, \forall x_{46} < H\, F_{29}$	$\forall I$ 123
125	Γ_{11}	$(x_{44} < H \rightarrow \forall x_{45} < H\, \forall x_{46} < H\, F_{29})$	$\rightarrow I$ 124
126	Γ_{11}	$\forall x_{44} < H\, \forall x_{45} < H\, \forall x_{46} < H\, F_{29}$	$\forall I$ 125
127	Γ_{16}	F_{30}	$\rightarrow I$ 126
128	Γ_{17}	F_{30}	Closed-Weak 127
129	Γ_{0}	$\forall x_{18} < (n+1)\, \exists x_{19} < H\, F_{31}$	$\wedge E_1$ 4
130	Γ_{18}	$\forall x_{18} < (n+1)\, \exists x_{19} < H\, F_{31}$	Weak 129
131	Γ_{17}	$\forall x_{18} < (n+1)\, \exists x_{19} < H\, F_{31}$	Weak 130
132	Γ_{19}	$\forall x_{18} < (n+1)\, \exists x_{19} < H\, F_{31}$	Weak 131
133	Γ_{20}	$\forall x_{18} < (n+1)\, \exists x_{19} < H\, F_{31}$	Weak 132
134	Γ_{21}	$\forall x_{18} < (n+1)\, \exists x_{19} < H\, F_{31}$	Weak 133
135	Γ_{22}	$\forall x_{18} < (n+1)\, \exists x_{19} < H\, F_{31}$	Weak 134
136	Γ_{23}	$\forall x_{18} < (n+1)\, \exists x_{19} < H\, F_{31}$	Weak 135
137	Γ_{23}	$(i < (n+1) \rightarrow F_{32})$	$\forall E$ 136
138	Γ_{17}	$((i < n \wedge n < (n+1)) \rightarrow i < (n+1))$	Axiom O2
139	Γ_{16}	F_{33}	Lemma 29
140	Γ_{17}	F_{33}	Closed-Weak 139
141	Γ_{17}	$\forall x_{53}\, \forall x_{54}\, ((i + x_{54}) < (x_{53} + x_{54}) \rightarrow i < x_{53})$	$\forall E$ 140
142	Γ_{17}	$\forall x_{54}\, ((i + x_{54}) < (n + x_{54}) \rightarrow i < n)$	$\forall E$ 141
143	Γ_{17}	$((i+1) < (n+1) \rightarrow i < n)$	$\forall E$ 142
144	Γ_{17}	$(i+1) < (n+1)$	Assume
145	Γ_{17}	$i < n$	$\rightarrow E$ 143,144
146	Γ_{16}	$\forall x_{55}\, x_{55} < (x_{55} + 1)$	Lemma 2
147	Γ_{17}	$\forall x_{55}\, x_{55} < (x_{55} + 1)$	Closed-Weak 146
148	Γ_{17}	$n < (n+1)$	$\forall E$ 147
149	Γ_{17}	$(i < n \wedge n < (n+1))$	$\wedge I$ 145,148
150	Γ_{17}	$i < (n+1)$	$\rightarrow E$ 138,149
151	Γ_{19}	$i < (n+1)$	Weak 150
152	Γ_{20}	$i < (n+1)$	Weak 151
153	Γ_{21}	$i < (n+1)$	Weak 152
154	Γ_{22}	$i < (n+1)$	Weak 153
155	Γ_{23}	$i < (n+1)$	Weak 154
156	Γ_{23}	F_{32}	$\rightarrow E$ 137,155
157	Γ_{24}	F_{4}	Assume
158	Γ_{24}	$\exists x_{20} < H\, \exists x_{21} < H\, \exists x_{22} < H\, \mathsf{At}(i, x_{40}, x_{20}, x_{21}, x_{22})$	$\wedge E_2$ 157
159	Γ_{25}	$(x_{41} < H \wedge \exists x_{21} < H\, \exists x_{22} < H\, \mathsf{At}(i, x_{40}, x_{41}, x_{21}, x_{22}))$	Assume
160	Γ_{25}	$\exists x_{21} < H\, \exists x_{22} < H\, \mathsf{At}(i, x_{40}, x_{41}, x_{21}, x_{22})$	$\wedge E_2$ 159
161	Γ_{26}	$(x_{42} < H \wedge \exists x_{22} < H\, \mathsf{At}(i, x_{40}, x_{41}, x_{42}, x_{22}))$	Assume
162	Γ_{26}	$\exists x_{22} < H\, \mathsf{At}(i, x_{40}, x_{41}, x_{42}, x_{22})$	$\wedge E_2$ 161
163	Γ_{0}	Close	$\wedge E_1$ 2
164	Γ_{18}	Close	Weak 163
165	Γ_{17}	Close	Weak 164
166	Γ_{19}	Close	Weak 165
167	Γ_{20}	Close	Weak 166

Line	Open assumptions	Formula	Rule and premises
168	Γ_{21}	Close	Weak 167
169	Γ_{22}	Close	Weak 168
170	Γ_{23}	Close	Weak 169
171	Γ_{24}	Close	Weak 170
172	Γ_{25}	Close	Weak 171
173	Γ_{26}	Close	Weak 172
174	Γ_{27}	Close	Weak 173
175	Γ_{27}	$\forall x_6 < H\, \forall x_7 < H\, \forall x_8 < H\, \forall x_9 < H\, \forall x_{10} < H\, F_{35}$	$\forall E$ 174
176	Γ_{27}	$(x_{40} < H \to \forall x_7 < H\, \forall x_8 < H\, \forall x_9 < H\, \forall x_{10} < H\, F_{37})$	$\forall E$ 175
177	Γ_{24}	$x_{40} < H$	$\wedge E_1$ 157
178	Γ_{25}	$x_{40} < H$	Weak 177
179	Γ_{26}	$x_{40} < H$	Weak 178
180	Γ_{27}	$x_{40} < H$	Weak 179
181	Γ_{27}	$\forall x_7 < H\, \forall x_8 < H\, \forall x_9 < H\, \forall x_{10} < H\, F_{37}$	$\to E$ 176,180
182	Γ_{27}	$(x_{41} < H \to \forall x_8 < H\, \forall x_9 < H\, \forall x_{10} < H\, F_{39})$	$\forall E$ 181
183	Γ_{25}	$x_{41} < H$	$\wedge E_1$ 159
184	Γ_{26}	$x_{41} < H$	Weak 183
185	Γ_{27}	$x_{41} < H$	Weak 184
186	Γ_{27}	$\forall x_8 < H\, \forall x_9 < H\, \forall x_{10} < H\, F_{39}$	$\to E$ 182,185
187	Γ_{27}	$(x_{42} < H \to \forall x_9 < H\, \forall x_{10} < H\, F_{41})$	$\forall E$ 186
188	Γ_{26}	$x_{42} < H$	$\wedge E_1$ 161
189	Γ_{27}	$x_{42} < H$	Weak 188
190	Γ_{27}	$\forall x_9 < H\, \forall x_{10} < H\, F_{41}$	$\to E$ 187,189
191	Γ_{27}	$(x_{43} < H \to \forall x_{10} < H\, F_{43})$	$\forall E$ 190
192	Γ_{27}	$(x_{43} < H \wedge \mathsf{At}(i, x_{40}, x_{41}, x_{42}, x_{43}))$	Assume
193	Γ_{27}	$x_{43} < H$	$\wedge E_1$ 192
194	Γ_{27}	$\forall x_{10} < H\, F_{43}$	$\to E$ 191,193
195	Γ_{27}	$(x_{48} < H \to F_{45})$	$\forall E$ 194
196	Γ_{19}	$x_{48} < H$	Assume
197	Γ_{20}	$x_{48} < H$	Weak 196
198	Γ_{21}	$x_{48} < H$	Weak 197
199	Γ_{22}	$x_{48} < H$	Weak 198
200	Γ_{23}	$x_{48} < H$	Weak 199
201	Γ_{24}	$x_{48} < H$	Weak 200
202	Γ_{25}	$x_{48} < H$	Weak 201
203	Γ_{26}	$x_{48} < H$	Weak 202
204	Γ_{27}	$x_{48} < H$	Weak 203
205	Γ_{27}	F_{45}	$\to E$ 195,204
206	Γ_{27}	$(x_{49} < H \to \forall x_{12} < H\, \forall x_{13} < H\, (F_{46} \to I((i+1), x_{48}, x_{49}, x_{12}, x_{13})))$	$\forall E$ 205
207	Γ_{20}	$x_{49} < H$	Assume
208	Γ_{21}	$x_{49} < H$	Weak 207
209	Γ_{22}	$x_{49} < H$	Weak 208
210	Γ_{23}	$x_{49} < H$	Weak 209
211	Γ_{24}	$x_{49} < H$	Weak 210
212	Γ_{25}	$x_{49} < H$	Weak 211
213	Γ_{26}	$x_{49} < H$	Weak 212
214	Γ_{27}	$x_{49} < H$	Weak 213
215	Γ_{27}	$\forall x_{12} < H\, \forall x_{13} < H\, (F_{46} \to I((i+1), x_{48}, x_{49}, x_{12}, x_{13}))$	$\to E$ 206,214
216	Γ_{27}	$(x_{50} < H \to \forall x_{13} < H\, (F_{47} \to I((i+1), x_{48}, x_{49}, x_{50}, x_{13})))$	$\forall E$ 215
217	Γ_{21}	$x_{50} < H$	Assume
218	Γ_{22}	$x_{50} < H$	Weak 217

Line	Open assumptions	Formula	Rule and premises
219	Γ_{23}	$x_{50} < H$	Weak 218
220	Γ_{24}	$x_{50} < H$	Weak 219
221	Γ_{25}	$x_{50} < H$	Weak 220
222	Γ_{26}	$x_{50} < H$	Weak 221
223	Γ_{27}	$x_{50} < H$	Weak 222
224	Γ_{27}	$\forall x_{13} < H\,(F_{47} \to I((i+1), x_{48}, x_{49}, x_{50}, x_{13}))$	$\to E$ 216,223
225	Γ_{27}	$(x_{51} < H \to (F_{48} \to I((i+1), x_{48}, x_{49}, x_{50}, x_{51})))$	$\forall E$ 224
226	Γ_{22}	$x_{51} < H$	Assume
227	Γ_{23}	$x_{51} < H$	Weak 226
228	Γ_{24}	$x_{51} < H$	Weak 227
229	Γ_{25}	$x_{51} < H$	Weak 228
230	Γ_{26}	$x_{51} < H$	Weak 229
231	Γ_{27}	$x_{51} < H$	Weak 230
232	Γ_{27}	$(F_{48} \to I((i+1), x_{48}, x_{49}, x_{50}, x_{51}))$	$\to E$ 225,231
233	Γ_{18}	$\varphi(i)$	Assume
234	Γ_{17}	$\varphi(i)$	Weak 233
235	Γ_{19}	$\varphi(i)$	Weak 234
236	Γ_{20}	$\varphi(i)$	Weak 235
237	Γ_{21}	$\varphi(i)$	Weak 236
238	Γ_{22}	$\varphi(i)$	Weak 237
239	Γ_{23}	$\varphi(i)$	Weak 238
240	Γ_{24}	$\varphi(i)$	Weak 239
241	Γ_{25}	$\varphi(i)$	Weak 240
242	Γ_{26}	$\varphi(i)$	Weak 241
243	Γ_{27}	$\varphi(i)$	Weak 242
244	Γ_{24}	$i < (n+1)$	Weak 155
245	Γ_{25}	$i < (n+1)$	Weak 244
246	Γ_{26}	$i < (n+1)$	Weak 245
247	Γ_{27}	$i < (n+1)$	Weak 246
248	Γ_{27}	$\forall x_{44} < H\,\forall x_{45} < H\,F_{13}$	$\to E$ 243,247
249	Γ_{27}	$(x_{40} < H \to \forall x_{45} < H\,F_{49})$	$\forall E$ 248
250	Γ_{27}	$\forall x_{45} < H\,F_{49}$	$\to E$ 249,180
251	Γ_{27}	$(x_{41} < H \to F_{50})$	$\forall E$ 250
252	Γ_{27}	F_{50}	$\to E$ 251,185
253	Γ_{27}	F_{51}	$\forall E$ 252
254	Γ_{27}	$\forall x_{47} < H\,(\mathsf{At}(i, x_{40}, x_{41}, x_{42}, x_{47}) \to I(i, x_{40}, x_{41}, x_{42}, x_{47}))$	$\to E$ 253,189
255	Γ_{27}	$(x_{43} < H \to (\mathsf{At}(i, x_{40}, x_{41}, x_{42}, x_{43}) \to I(i, x_{40}, x_{41}, x_{42}, x_{43})))$	$\forall E$ 254
256	Γ_{27}	$(\mathsf{At}(i, x_{40}, x_{41}, x_{42}, x_{43}) \to I(i, x_{40}, x_{41}, x_{42}, x_{43}))$	$\to E$ 255,193
257	Γ_{27}	$\mathsf{At}(i, x_{40}, x_{41}, x_{42}, x_{43})$	$\wedge E_2$ 192
258	Γ_{27}	$I(i, x_{40}, x_{41}, x_{42}, x_{43})$	$\to E$ 256,257
259	Γ_0	$\forall x_{27} < n\,F_8$	$\wedge E_1$ 6
260	Γ_{18}	$\forall x_{27} < n\,F_8$	Weak 259
261	Γ_{17}	$\forall x_{27} < n\,F_8$	Weak 260
262	Γ_{19}	$\forall x_{27} < n\,F_8$	Weak 261
263	Γ_{20}	$\forall x_{27} < n\,F_8$	Weak 262
264	Γ_{21}	$\forall x_{27} < n\,F_8$	Weak 263
265	Γ_{22}	$\forall x_{27} < n\,F_8$	Weak 264
266	Γ_{23}	$\forall x_{27} < n\,F_8$	Weak 265
267	Γ_{24}	$\forall x_{27} < n\,F_8$	Weak 266
268	Γ_{25}	$\forall x_{27} < n\,F_8$	Weak 267
269	Γ_{26}	$\forall x_{27} < n\,F_8$	Weak 268

Line	Open assumptions	Formula	Rule and premises
270	Γ_{27}	$\forall x_{27} < n\, F_8$	Weak 269
271	Γ_{27}	$(i < n \to \forall x_{28} < H\, \forall x_{29} < H\, \forall x_{30} < H\, F_{53})$	$\forall E$ 270
272	Γ_{19}	$i < n$	Weak 145
273	Γ_{20}	$i < n$	Weak 272
274	Γ_{21}	$i < n$	Weak 273
275	Γ_{22}	$i < n$	Weak 274
276	Γ_{23}	$i < n$	Weak 275
277	Γ_{24}	$i < n$	Weak 276
278	Γ_{25}	$i < n$	Weak 277
279	Γ_{26}	$i < n$	Weak 278
280	Γ_{27}	$i < n$	Weak 279
281	Γ_{27}	$\forall x_{28} < H\, \forall x_{29} < H\, \forall x_{30} < H\, F_{53}$	$\to E$ 271,280
282	Γ_{27}	$(x_{40} < H \to \forall x_{29} < H\, \forall x_{30} < H\, F_{55})$	$\forall E$ 281
283	Γ_{27}	$\forall x_{29} < H\, \forall x_{30} < H\, F_{55}$	$\to E$ 282,180
284	Γ_{27}	$(x_{41} < H \to \forall x_{30} < H\, F_{57})$	$\forall E$ 283
285	Γ_{27}	$\forall x_{30} < H\, F_{57}$	$\to E$ 284,185
286	Γ_{27}	$(x_{42} < H \to F_{59})$	$\forall E$ 285
287	Γ_{27}	F_{59}	$\to E$ 286,189
288	Γ_{27}	$(x_{43} < H \to \forall x_{32} < H\, \forall x_{33} < H\, \forall x_{34} < H\, \forall x_{35} < H\, F_{60})$	$\forall E$ 287
289	Γ_{27}	$\forall x_{32} < H\, \forall x_{33} < H\, \forall x_{34} < H\, \forall x_{35} < H\, F_{60}$	$\to E$ 288,193
290	Γ_{27}	$(x_{48} < H \to \forall x_{33} < H\, \forall x_{34} < H\, \forall x_{35} < H\, F_{61})$	$\forall E$ 289
291	Γ_{27}	$\forall x_{33} < H\, \forall x_{34} < H\, \forall x_{35} < H\, F_{61}$	$\to E$ 290,204
292	Γ_{27}	$(x_{49} < H \to \forall x_{34} < H\, \forall x_{35} < H\, F_{62})$	$\forall E$ 291
293	Γ_{27}	$\forall x_{34} < H\, \forall x_{35} < H\, F_{62}$	$\to E$ 292,214
294	Γ_{27}	$(x_{50} < H \to \forall x_{35} < H\, F_{63})$	$\forall E$ 293
295	Γ_{27}	$\forall x_{35} < H\, F_{63}$	$\to E$ 294,223
296	Γ_{27}	$(x_{51} < H \to F_{64})$	$\forall E$ 295
297	Γ_{27}	F_{64}	$\to E$ 296,231
298	Γ_{23}	$\mathsf{At}((i+1), x_{48}, x_{49}, x_{50}, x_{51})$	Assume
299	Γ_{24}	$\mathsf{At}((i+1), x_{48}, x_{49}, x_{50}, x_{51})$	Weak 298
300	Γ_{25}	$\mathsf{At}((i+1), x_{48}, x_{49}, x_{50}, x_{51})$	Weak 299
301	Γ_{26}	$\mathsf{At}((i+1), x_{48}, x_{49}, x_{50}, x_{51})$	Weak 300
302	Γ_{27}	$\mathsf{At}((i+1), x_{48}, x_{49}, x_{50}, x_{51})$	Weak 301
303	Γ_{27}	$(\mathsf{At}(i, x_{40}, x_{41}, x_{42}, x_{43}) \wedge \mathsf{At}((i+1), x_{48}, x_{49}, x_{50}, x_{51}))$	$\wedge I$ 257,302
304	Γ_{27}	$\mathsf{Step}(x_{40}, x_{41}, x_{42}, x_{43}, x_{48}, x_{49}, x_{50}, x_{51})$	$\to E$ 297,303
305	Γ_{27}	F_{48}	$\wedge I$ 258,304
306	Γ_{27}	$I((i+1), x_{48}, x_{49}, x_{50}, x_{51})$	$\to E$ 232,305
307	Γ_{26}	$I((i+1), x_{48}, x_{49}, x_{50}, x_{51})$	$\exists E$ 162,306
308	Γ_{25}	$I((i+1), x_{48}, x_{49}, x_{50}, x_{51})$	$\exists E$ 160,307
309	Γ_{24}	$I((i+1), x_{48}, x_{49}, x_{50}, x_{51})$	$\exists E$ 158,308
310	Γ_{23}	$I((i+1), x_{48}, x_{49}, x_{50}, x_{51})$	$\exists E$ 156,309
311	Γ_{22}	$(\mathsf{At}((i+1), x_{48}, x_{49}, x_{50}, x_{51}) \to I((i+1), x_{48}, x_{49}, x_{50}, x_{51}))$	$\to I$ 310
312	Γ_{21}	F_{65}	$\to I$ 311
313	Γ_{21}	F_3	$\forall I$ 312
314	Γ_{20}	$(x_{50} < H \to F_3)$	$\to I$ 313
315	Γ_{20}	$\forall x_{50} < H\, F_3$	$\forall I$ 314
316	Γ_{19}	$(x_{49} < H \to \forall x_{50} < H\, F_3)$	$\to I$ 315
317	Γ_{19}	$\forall x_{49} < H\, \forall x_{50} < H\, F_3$	$\forall I$ 316
318	Γ_{17}	$(x_{48} < H \to \forall x_{49} < H\, \forall x_{50} < H\, F_3)$	$\to I$ 317
319	Γ_{17}	$\forall x_{48} < H\, \forall x_{49} < H\, \forall x_{50} < H\, F_3$	$\forall I$ 318
320	Γ_{17}	$\forall x_{44} < H\, \forall x_{45} < H\, \forall x_{46} < H\, F_{29}$	$\to E$ 128,319

Line	Open assumptions	Formula	Rule and premises
321	Γ_{18}	$\varphi((i+1))$	$\to I$ 320
322	Γ_0	$(\varphi(i) \to \varphi((i+1)))$	$\to I$ 321
323	Γ_0	$\forall i\,(\varphi(i) \to \varphi((i+1)))$	$\forall I$ 322
324	Γ_0	$(\varphi(0) \wedge \forall i\,(\varphi(i) \to \varphi((i+1))))$	$\wedge I$ 95,323
325	Γ_0	$\forall i < (n+1)\,\forall x_{44} < H\,\forall x_{45} < H\,F_{13}$	$\to E$ 38,324
326	Γ_0	$\varphi(n)$	$\forall E$ 325
327	Γ_0	$\forall x_{55}\,x_{55} < (x_{55}+1)$	Closed-Weak 146
328	Γ_0	$n < (n+1)$	$\forall E$ 327
329	Γ_0	$\forall x_{44} < H\,\forall x_{45} < H\,F_{66}$	$\to E$ 326,328
330	Γ_1	$\forall x_{44} < H\,\forall x_{45} < H\,F_{66}$	Weak 329
331	Γ_2	$\forall x_{44} < H\,\forall x_{45} < H\,F_{66}$	Weak 330
332	Γ_3	$\forall x_{44} < H\,\forall x_{45} < H\,F_{66}$	Weak 331
333	Γ_4	$\forall x_{44} < H\,\forall x_{45} < H\,F_{66}$	Weak 332
334	Γ_4	$(x_{40} < H \to \forall x_{45} < H\,F_{67})$	$\forall E$ 333
335	Γ_4	$\forall x_{45} < H\,F_{67}$	$\to E$ 334,23
336	Γ_4	$(x_{41} < H \to F_{68})$	$\forall E$ 335
337	Γ_4	F_{68}	$\to E$ 336,28
338	Γ_4	F_{69}	$\forall E$ 337
339	Γ_4	$\forall x_{47} < H\,(\mathsf{At}(n, x_{40}, x_{41}, x_{42}, x_{47}) \to I(n, x_{40}, x_{41}, x_{42}, x_{47}))$	$\to E$ 338,32
340	Γ_4	$(x_{43} < H \to (\mathsf{At}(n, x_{40}, x_{41}, x_{42}, x_{43}) \to I(n, x_{40}, x_{41}, x_{42}, x_{43})))$	$\forall E$ 339
341	Γ_4	$(\mathsf{At}(n, x_{40}, x_{41}, x_{42}, x_{43}) \to I(n, x_{40}, x_{41}, x_{42}, x_{43}))$	$\to E$ 340,36
342	Γ_4	$(\mathsf{At}(n, x_{40}, x_{41}, x_{42}, x_{43}) \wedge x_{40} = 2)$	$\wedge E_2$ 35
343	Γ_4	$\mathsf{At}(n, x_{40}, x_{41}, x_{42}, x_{43})$	$\wedge E_1$ 342
344	Γ_4	$I(n, x_{40}, x_{41}, x_{42}, x_{43})$	$\to E$ 341,343
345	Γ_4	$x_{40} = 2$	$\wedge E_2$ 342
346	Γ_4	$(I(n, x_{40}, x_{41}, x_{42}, x_{43}) \wedge x_{40} = 2)$	$\wedge I$ 344,345
347	Γ_4	$g_0 < g_1$	$\to E$ 37,346
348	Γ_3	$g_0 < g_1$	$\exists E$ 13,347
349	Γ_2	$g_0 < g_1$	$\exists E$ 11,348
350	Γ_1	$g_0 < g_1$	$\exists E$ 9,349
351	Γ_0	$g_0 < g_1$	$\exists E$ 7,350
352	Γ_{16}	$((\mathsf{Init} \wedge (\mathsf{Close} \wedge (\mathsf{Final} \wedge \mathsf{Run}))) \to g_0 < g_1)$	$\to I$ 351

9 Reproducibility and references

The source archive includes the proof generators, exact JSON natural-deduction traces, finite certificates, the complete standalone Metamath database, and its verification log and SHA-256 manifest. Check the standalone file with an official Metamath executable by running `READ` on it, followed by `VERIFY PROOF *`. Require both a clean parse and a clean proof check: the presence of a final success message alone is not sufficient if an earlier error was printed.

The search and proof generator are untrusted. An invalid certificate or incorrect arithmetic computation must fail when its emitted proof is checked. The independent kernel check is what turns the finite computations into formal theorems. The finite reproduction of Rocq's gas-8 test is background evidence about the parameter only; it is not a premise of the final theorem.

1. T. Radó, "On non-computable functions," *Bell System Technical Journal* 41 (1962), 877–884. https://doi.org/10.1002/j.1538-7305.1962.tb00480.x.

2. Coq-BB5, BB2 enumeration, pipeline and verification sources: https://github.com/ccz181078/Coq-BB5/tree/main/CoqBB5/BB2. In particular `BB2_Deciders_Pipeline.v`, `Decider_Loop.v` and `TNF.v`. Exact retrieved source files are included in the archive.

3. Published BB2 certificate list: https://docs.bbchallenge.org/CoqBB5_release_v1.0.0/BB2_verified_enumeration.csv.

4. N. Megill and D. A. Wheeler, *Metamath: A Computer Language for Mathematical Proofs*. https://us.metamath.org/downloads/metamath.pdf. This project uses its own small first-order arithmetic database; it does not import set theory from `set.mm`.

5. P. Hájek and P. Pudlák, *Metamathematics of First-Order Arithmetic*, Springer, 1993. Background on bounded formulas and weak arithmetic; no theorem from this book is used as an extra axiom here.